

Rental Management

Functional Requirements and Dashboard Specifications

Module 7

Professional formatted edition

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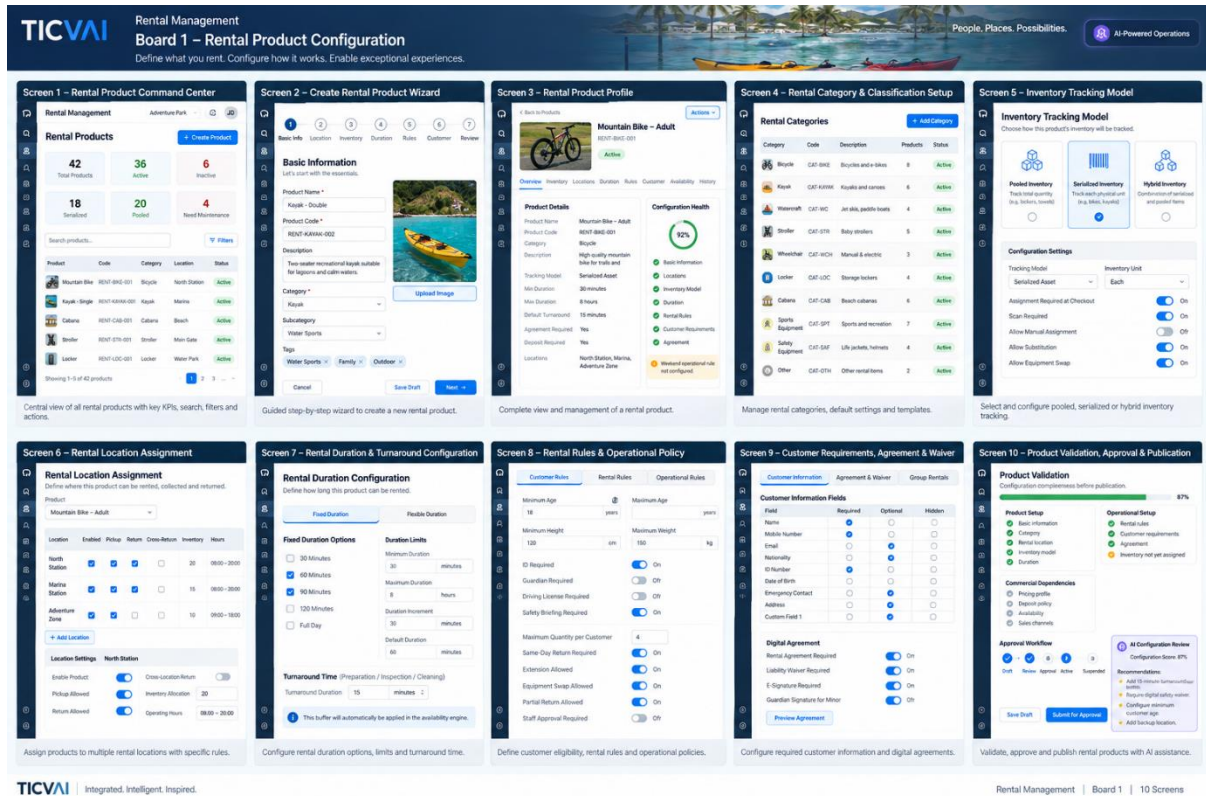
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Board 1 — Rental Product Configuration

Module: 7 — Rental Management

Board Objective: Create and configure the master definition of every rentable product before inventory, availability, pricing, booking, and operational rules are applied.



Board 1 should answer one fundamental question:

“What is being rented, where can it be rented, how is it tracked, and what are the fundamental operational rules of that rental product?”

The source requires rental products such as bikes, kayaks, watercraft, strollers, cabanas, lockers and wheelchairs, with product codes, categories, locations, inventory quantities, rental durations, rules, seasonal availability, blackout dates and maintenance periods.

I have also incorporated our recommended **pooled / serialized / hybrid inventory architecture**, turnaround time, waiver requirements, operational controls and AI assistance.

Screen 1 — Rental Product Command Center

Purpose

Central management screen for all rental products across tenants, venues and rental locations.

Functional Scope

Display configurable KPI cards:

- Total Rental Products
- Active Products
- Inactive Products
- Serialized Products
- Pooled Products
- Products Requiring Maintenance
- Products With Configuration Issues

- Products Awaiting Approval

Product table shall display:

Field	Example
Product	Mountain Bike
Code	RENT-BIKE-001
Category	Bicycle
Venue	Adventure Park
Rental Location	North Rental Station
Tracking	Serialized
Inventory	48
Availability	Available
Status	Active
Configuration	Complete

Support filters by:

- Tenant
- Venue
- Rental location
- Category
- Product status
- Tracking model
- Availability status
- Season
- Configuration status

Actions

+ Create Rental Product

Additional actions:

- Edit
- Duplicate
- Activate/Deactivate
- Archive
- View Inventory
- View Availability
- View Rental History
- Submit for Approval

AI Recommendation

Example:

AI Configuration Alert

Kayak – Double has inventory assigned but no turnaround time configured. Recommended turnaround: 15 minutes.

Screen 2 — Create Rental Product Wizard

Purpose

Guided workflow for creating a new rental product.

Wizard

Step 1 — Basic Information

- Product Name
- Product Code / SKU
- Internal Name
- Description
- Category
- Subcategory
- Product Image
- Tags
- Rental Type

Step 2 — Location

- Tenant
- Venue
- Rental Location(s)

Step 3 — Inventory Model

- Quantity / Pooled
- Serialized Asset
- Hybrid

Step 4 — Rental Duration

Step 5 — Operational Rules

Step 6 — Customer Requirements

Step 7 — Review & Publish

The system shall provide configuration validation before publication.

Example:

- ⚠ Product cannot be published.
Rental location has not been configured.

Support:

Save Draft | Previous | Next | Submit for Approval

Screen 3 — Rental Product Profile

Purpose

Provide the complete master configuration of a rental product.

Product Header

Mountain Bike — Adult

RENT-BIKE-001

Status:

ACTIVE

Display tabs:

Overview | Inventory | Locations | Duration | Rules | Customer Requirements | Availability | History

Overview

Show:

- Product name
- Code
- Category
- Description
- Images
- Tracking model
- Inventory quantity
- Rental locations
- Minimum duration
- Maximum duration
- Default turnaround time
- Rental agreement requirement
- Deposit requirement
- Current status

Configuration Health

Example:

Configuration completeness: 92%

Missing:

Weekend operational rule has not been configured.

The product profile becomes the master reference used by downstream Rental Management functions.

Screen 4 — Rental Category & Classification Setup

Purpose

Configure reusable categories rather than hard-coding individual rental types.

Examples:

- Bicycle
- Kayak
- Watercraft
- Stroller
- Wheelchair
- Locker
- Cabana
- Sports Equipment
- Safety Equipment
- Other Equipment

Each category shall support:

- Category code

- Category name
- Description
- Parent category
- Icon/Image
- Default inventory model
- Default rental duration
- Default turnaround time
- Default customer requirements
- Default rental rules
- Default waiver requirement
- Default deposit behavior

Templates

Administrators shall be able to create reusable:

Rental Product Templates

Example:

Water Sports Equipment Template

Automatically inherit:

- ID requirement
- Digital waiver
- Safety acknowledgment
- 15-minute turnaround
- Deposit requirement

Individual products can override permitted template settings.

Screen 5 — Inventory Tracking Model

This is one of the important enhancements I recommend adding to the original requirements.

Purpose

Determine **how the product's physical inventory will be controlled**.

Tracking Model

Option A — Pooled Inventory

Example:

Lockers

Total Quantity: 150

The booking engine reserves quantity but does not require a specific physical unit at booking.

Option B — Serialized Inventory

Example:

Bicycles

Every physical asset receives a unique identifier:

- BIKE-001
- BIKE-002
- BIKE-003

Serial numbers are assigned and managed in **Board 2**.
The original requirements specifically call for unique serial-number assignment and complete rental history by serial number.

Option C — Hybrid Inventory

Example:

Kayak Package

Rental product:

Double Kayak Package

May include:

- 1 serialized kayak
- 2 pooled paddles
- 2 pooled life jackets

- Configuration**
- Tracking Model
 - Inventory Unit
 - Quantity Control
 - Assignment Required at Checkout
 - Scan Required
 - Allow Manual Assignment
 - Allow Substitution
 - Allow Equipment Swap

This architecture will give TICVAI much more flexibility than supporting serial numbers alone.

Screen 6 — Rental Location Assignment

Purpose

Define where a product can physically be rented, collected and returned.

Support:

- Multiple venues
- Multiple rental stations
- Different pickup locations
- Different return locations

Example:

Mountain Bike

Location	Available	Pickup	Return
North Station	✓	✓	✓
Marina Station	✓	✓	✓
Adventure Zone	✓	✓	—

- Configuration**
- For each location:
- Enable/Disable Product

- Inventory Allocation
- Pickup Allowed
- Return Allowed
- Cross-Location Return Allowed
- Operating Hours
- Location-specific rules
- Location-specific inventory buffer

Support:

Allow return to different location: YES/NO

The source specifically requires multiple rental locations and inventory transfers between locations.

Actual inventory transfer operations will be handled in **Board 2**.

Screen 7 — Rental Duration & Turnaround Configuration

Purpose

Define how long the product can be rented.

The source requires minimum and maximum rental durations and both fixed and customer-defined durations.

Duration Type

Fixed Duration

Examples:

- 30 minutes
- 60 minutes
- 90 minutes
- 120 minutes
- Full Day

Flexible Duration

Customer chooses duration within configured limits.

Configuration

- Minimum Duration
- Maximum Duration
- Duration Increment
- Default Duration
- Allow Extension
- Maximum Extension
- Same-Day Return Required
- Overnight Rental Allowed

Recommended Addition — Turnaround Time

Configure:

Preparation / Inspection / Cleaning Buffer

Example:

Rental:

10:00 → 11:00

Turnaround:

11:00 → 11:15

Next available booking:

11:15

This buffer must automatically feed into the availability engine in Board 3.

Screen 8 — Rental Rules & Operational Policy

Purpose

Define the fundamental operational restrictions for the product.

Rules may include:

Customer Rules

- Minimum Age
- Maximum Age
- Minimum Height
- Maximum Weight
- ID Required
- Guardian Required
- Membership Required
- Driving License Required
- Safety Briefing Required

Rental Rules

- Maximum Quantity per Customer
- Same-Day Return Required
- Return Location Restriction
- Extension Allowed
- Equipment Swap Allowed
- Partial Return Allowed
- Staff Approval Required

Operational Rules

- Checkout Scan Required
- Return Scan Required
- Condition Inspection Required
- Photo at Checkout Required
- Photo at Return Required

Rules should support:

Required | Optional | Not Applicable

Detailed damage, return and exception workflows will be configured in later boards.

Screen 9 — Customer Requirements, Agreement & Waiver

This formally incorporates another recommendation from requirement 7.6.15 into the actual product configuration.

Purpose

Define information and agreements required before a customer can receive the rental.

Customer Information

Configurable fields:

- Name
- Mobile
- Email
- Nationality
- ID Number
- Date of Birth
- Emergency Contact
- Address
- Custom Fields

Fields can be configured:

Required / Optional / Hidden

The source already calls for configurable customer and participant information for rental bookings.

Digital Agreement

Configure:

- Rental Agreement Required
- Liability Waiver Required
- Terms & Conditions
- Safety Declaration
- E-Signature Required
- Guardian Signature for Minor
- Agreement Version

Group Rentals

Support:

- Primary renter
- Participant details
- Individual waiver
- Group waiver
- Guardian consent

The signed agreement must be linked to the booking and retained according to the client's configured retention policy.

Screen 10 — Product Validation, Approval & Publication

Purpose

Provide controlled governance before a rental product becomes sellable.

Configuration Validation

System checks:

Product Setup

- ✓ Basic information
- ✓ Category

- ✓ Rental location
- ✓ Inventory model
- ✓ Duration

Operational Setup

- ✓ Rental rules
- ✓ Customer requirements
- ✓ Agreement
- ⚠ Inventory not yet assigned

Commercial Dependencies

- Pricing profile
- Deposit policy
- Availability
- Sales channels

Some dependencies may be completed in subsequent boards before the product becomes commercially available.

Workflow

Draft → Configuration Review → Approval → Active → Suspended → Archived

Support:

- Maker/checker workflow
- Approval comments
- Rejection reason
- Version history
- Effective date
- Future activation
- Suspension
- Re-publication

AI Configuration Review

Before submission, AI analyzes the setup.

Example:

AI Configuration Review

Product: Double Kayak

Configuration Score: **87%**

Recommendations:

- Add 15-minute turnaround buffer.
- Require digital safety waiver.
- Configure minimum customer age.
- Inventory is assigned at Marina A but no backup location is configured.

AI recommendations remain advisory unless the client configures specific checks as mandatory.

Board 1 — Key Business Flow

The board should visually follow:

Create Product → Classify → Select Inventory Model → Assign Location → Define Duration → Configure Rules → Define Customer Requirements → Validate → Approve → Publish

Once published, the product becomes available to downstream configuration:

Board 2 Inventory → Board 3 Availability → Board 4 Pricing & Deposits → Board 5 Booking → Board 6 Checkout

Board 1 Acceptance Criteria

Board 1 will be considered complete when TICVAI can:

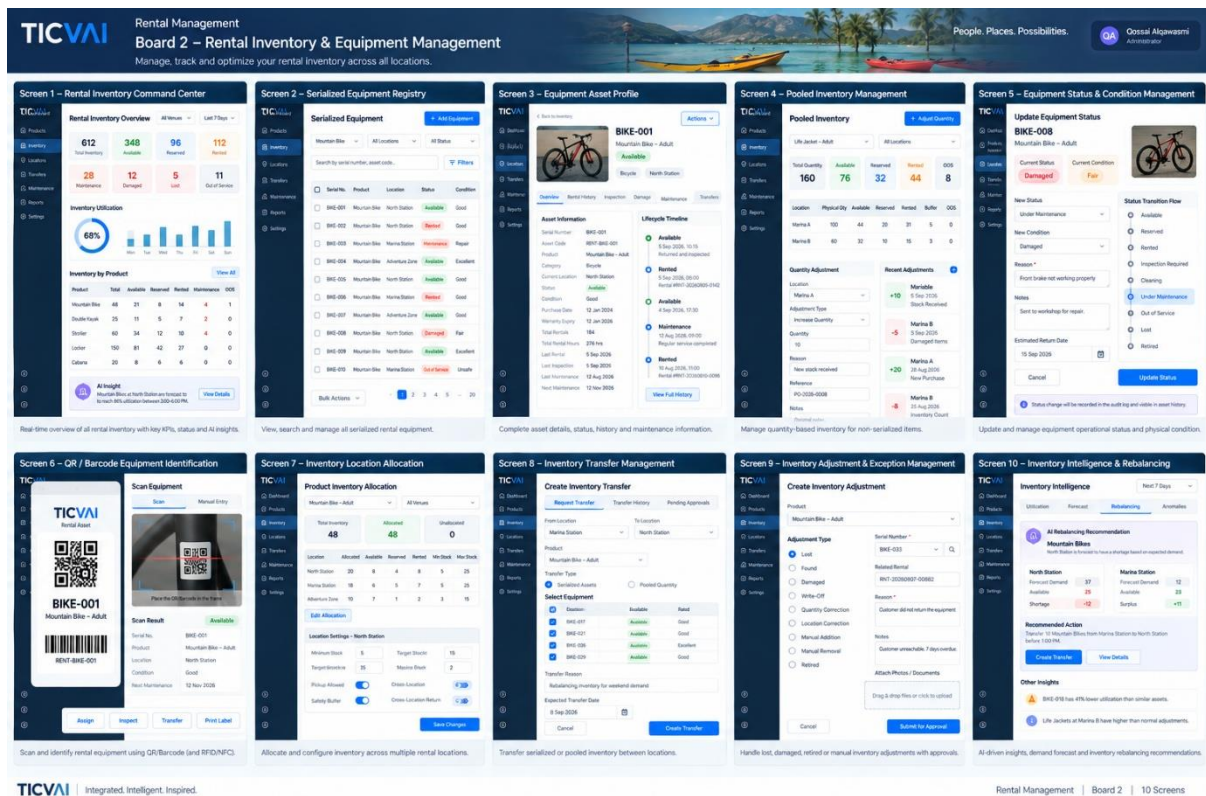
1. Create and maintain any type of rental product.
2. Support **pooled, serialized and hybrid inventory models**.
3. Associate products with one or multiple rental locations.
4. Configure fixed and flexible rental durations.
5. Configure minimum/maximum duration and turnaround buffers.
6. Define operational and customer eligibility requirements.
7. Configure digital agreements, waivers and required customer information.
8. Support reusable rental categories/templates.
9. Validate product configuration and manage approval/publication.
10. Maintain complete version and audit history.

This gives us a strong **master configuration layer** without duplicating the detailed inventory, availability, pricing or operational functionality that belongs in Boards 2–10.

Board 2 — Rental Inventory & Equipment Management

Module: 7 — Rental Management

Board Objective: Manage the physical rental inventory behind each rental product, including serialized assets, pooled inventory, equipment status, location allocation, transfers, scanning, history, and operational readiness.



Board 1 answered:

“What do we rent?”

Board 2 answers:

“What physical inventory do we actually have, where is it, what condition is it in, and can it currently be rented?”

The original requirement specifically requires unique serial numbers, activation/deactivation, statuses such as Available, Rented, Under Maintenance, Lost and Damaged, and complete rental history by serial number. It also requires real-time inventory, transfers between locations, maintenance blocking and out-of-service management.

I recommend **exactly 10 screens**.

Screen 1 — Rental Inventory Command Center

Purpose

Provide a real-time operational overview of all rental inventory across venues and locations.

KPI Cards

- Total Inventory
- Available Now
- Reserved
- Rented
- Under Maintenance
- Damaged

- Lost
- Out of Service
- Inventory Utilization %
- Inventory Alerts

Inventory Overview

Product	Total	Available	Reserved	Rented	Maintenance	OOS
Mountain Bike	48	21	8	14	4	1
Double Kayak	25	11	5	7	2	0
Stroller	60	34	12	10	4	0
Locker	150	81	42	27	0	0

Filters

- Tenant
- Venue
- Location
- Product
- Category
- Tracking Model
- Inventory Status
- Condition
- Availability

AI Insight

Potential shortage: Mountain Bikes at North Station are forecast to reach 96% utilization between 3:00–6:00 PM.

Actions:

View Inventory | Transfer Inventory | Add Equipment | Bulk Import

Screen 2 — Serialized Equipment Registry

Purpose

Manage individually identifiable rental assets.

Examples:

Bike #BIKE-001

Kayak #KAY-027

Wheelchair #WC-014

Equipment Table

Serial No.	Product	Location	Status	Condition	Last Rental
BIKE-001	Mountain Bike	North	Available	Good	07 Sep
BIKE-002	Mountain Bike	North	Rented	Good	Active
BIKE-003	Mountain Bike	Marina	Maintenance	Repair	05 Sep

Equipment Creation

Support:

- Manual creation
- Bulk creation
- CSV/import
- Barcode scanning
- QR scanning

Fields:

- Serial Number
- Asset Code
- Product
- Manufacturer Serial Number
- Location
- Acquisition Date
- Purchase Cost
- Warranty Expiry
- Condition
- Status

The system must prevent duplicate serial numbers.

Screen 3 — Equipment / Asset Profile

Purpose

Provide a complete digital record for every serialized rental asset.

Example:

BIKE-001

Mountain Bike — Adult

Status:

AVAILABLE

Asset Information

- Serial Number
- Asset Code
- Product
- Category
- Current Location
- Current Status
- Condition
- Purchase Date
- Warranty
- Total Rental Count
- Total Rental Hours
- Last Rental
- Last Inspection
- Last Maintenance
- Next Maintenance

Tabs

Overview | Rental History | Inspection | Damage | Maintenance | Transfers | Audit

Lifecycle Timeline

Example:

Purchased → Activated → Rented × 184 → Repaired → Returned to Service

The source requires complete rental history by serial number.

Screen 4 — Pooled Inventory Management

Purpose

Manage products where individual physical items do not require serial-level tracking.

Examples:

- Lockers
- Towels
- Helmets
- Paddles
- Life Jackets
- Other accessories

Inventory View

Product: Life Jacket – Adult

Location	Physical Qty	Available	Reserved	Rented	Buffer	OOS
Marina A	100	44	20	31	5	0
Marina B	60	32	10	15	3	0

Actions

- Increase Quantity
- Decrease Quantity
- Adjust Inventory
- Transfer Quantity
- Mark Out of Service

Every manual adjustment must require:

- Quantity
- Reason
- Staff Member
- Date/Time
- Optional Notes

and create an audit record.

Screen 5 — Equipment Status & Condition Management

Purpose

Separate **operational status** from **physical condition**.

This distinction is important.

Operational Status

- Available
- Reserved
- Rented
- Inspection Required
- Cleaning
- Under Maintenance
- Out of Service
- Lost
- Retired

Condition

- Excellent
- Good
- Fair
- Damaged
- Major Damage
- Unsafe

Example:

BIKE-028

Status:

Inspection Required

Condition:

Fair

Status Transition Rules

TICVAI should control valid transitions.

For example:

Rented → Returned → Inspection → Available

or:

Rented → Returned Damaged → Maintenance → Available

The system should prevent an unsafe/damaged asset from accidentally returning to rentable inventory.

Screen 6 — QR / Barcode Equipment Identification

Purpose

Give every physical serialized rental item a scannable digital identity.

This incorporates the equipment-assignment scanning recommendation from the source.

Supported Identifiers

- QR
- Barcode
- RFID where applicable
- NFC where applicable

Equipment Label

Example:

TICVAI Rental Asset

BIKE-001

[QR CODE]

Scan Result

When staff scans equipment:

BIKE-001 — Mountain Bike

Location: North Station

Status: Available

Condition: Good

Next Maintenance: 82 rental hours

Actions:

Assign | Inspect | Transfer | Maintenance | View History

Actual assignment to a customer reservation occurs in **Board 6 – Checkout & Equipment Assignment**.

Screen 7 — Inventory Location Allocation

Purpose

Manage how inventory is distributed among rental locations.

Example:

Mountain Bike — 48 Units

Location	Allocated	Available	Reserved	Rented
North Station	20	8	4	8
Marina Station	18	6	5	7
Adventure Zone	10	7	1	2

Configuration

For each location:

- Minimum Stock
- Target Stock
- Maximum Stock
- Safety Buffer
- Rental Allocation
- Reservation Allocation

Alerts

⚠ North Station inventory has fallen below minimum stock.

This screen manages physical inventory distribution. Board 3 will use that inventory to calculate future bookable availability.

Screen 8 — Inventory Transfer Management

Purpose

Control physical inventory movement between locations.

The original requirements explicitly include inventory transfers between locations.

Create Transfer

From: Marina Station

To: North Station

Product:

Mountain Bike

Select:

BIKE-017

BIKE-021

BIKE-026

Transfer Workflow

Requested → Approved → In Transit → Received → Completed

Support:

- Serialized asset transfer
- Pooled quantity transfer
- Bulk transfer
- Scheduled transfer
- Emergency transfer

Receiving

Receiving operator scans equipment.

System verifies:

Expected: 10

Received: 9

Alert:

⚠ BIKE-026 has not been received.

A discrepancy workflow must be generated.

Screen 9 — Inventory Adjustment & Exception Management

Purpose

Handle physical inventory discrepancies and exceptional conditions.

Adjustment Types

- Lost
- Found
- Damaged
- Write-Off
- Quantity Correction
- Stock Count Adjustment
- Manual Addition
- Manual Removal
- Retired

- Location Correction

Example

Asset: BIKE-033

Current Status: Available

Change to:

Lost

Reason:

Equipment not returned by customer

Related Rental:

RNT-20260907-00882

Approval Rules

High-risk adjustments can require supervisor approval.

Examples:

- Lost Asset
- Asset Write-Off
- Large pooled quantity adjustment
- Manual status override

Every adjustment must retain:

Before Value → After Value → Reason → User → Approval → Timestamp

Screen 10 — Inventory Intelligence & Rebalancing

Purpose

Use operational data and AI to improve rental inventory distribution and utilization.

This is one of the areas where TICVAI should go beyond the original matrix.

Inventory Health

Display:

- Utilization by product
- Utilization by location
- Idle equipment
- Shortage risk
- Excess inventory
- Maintenance impact
- Unavailable inventory
- Inventory turnover

AI Rebalancing Recommendation

Example:

Mountain Bikes

North Station

Forecast demand tomorrow: **37 units**

Forecast available inventory: **25 units**

Marina Station

Forecast demand: **12 units**

Forecast available inventory: **23 units**

AI Recommendation:

Transfer 10 Mountain Bikes from Marina Station to North Station before 1:00 PM.

Actions:

Review Recommendation | Create Transfer | Dismiss

AI Asset Insight

BIKE-018 has 41% lower utilization than similar assets because it has repeatedly been marked “Fair Condition.” Recommend inspection.

AI Inventory Anomaly

Physical inventory adjustment frequency for Adult Life Jackets at Marina B is significantly above normal.

AI remains advisory and should not automatically move or write off physical inventory without the appropriate authorization.

Board 2 — Inventory Lifecycle

The board should visually demonstrate:

Create Inventory → Identify/Serialize → Allocate Location → Monitor Status → Scan → Transfer → Adjust → Maintain History → Optimize

It should support both:

Serialized Asset

Product → BIKE-001 → Status → Location → Rental History
and:

Pooled Inventory

Product → Location → Quantity → Availability

This distinction is essential because the original requirements combine serial-number management with quantity-based real-time inventory.

Integration Boundaries

To avoid duplication with our other TICVAI modules:

Rental Board 2 owns:

Physical rental inventory, serial numbers, rental equipment status, rental locations, transfers and asset-level rental history.

Inventory & Procurement Module owns:

Procurement, suppliers, purchase orders, receiving and broader enterprise stock processes.

Rental Board 9 owns:

Detailed preventive/corrective maintenance scheduling and repair workflows.

Rental Board 6 owns:

Customer checkout and assignment of an asset to an actual rental.

Rental Board 8 owns:

Return inspection, damage assessment and customer settlement.

This separation is important so Softlabs doesn't build the same functionality multiple times.

Board 2 — Acceptance Criteria

Board 2 is complete when TICVAI can:

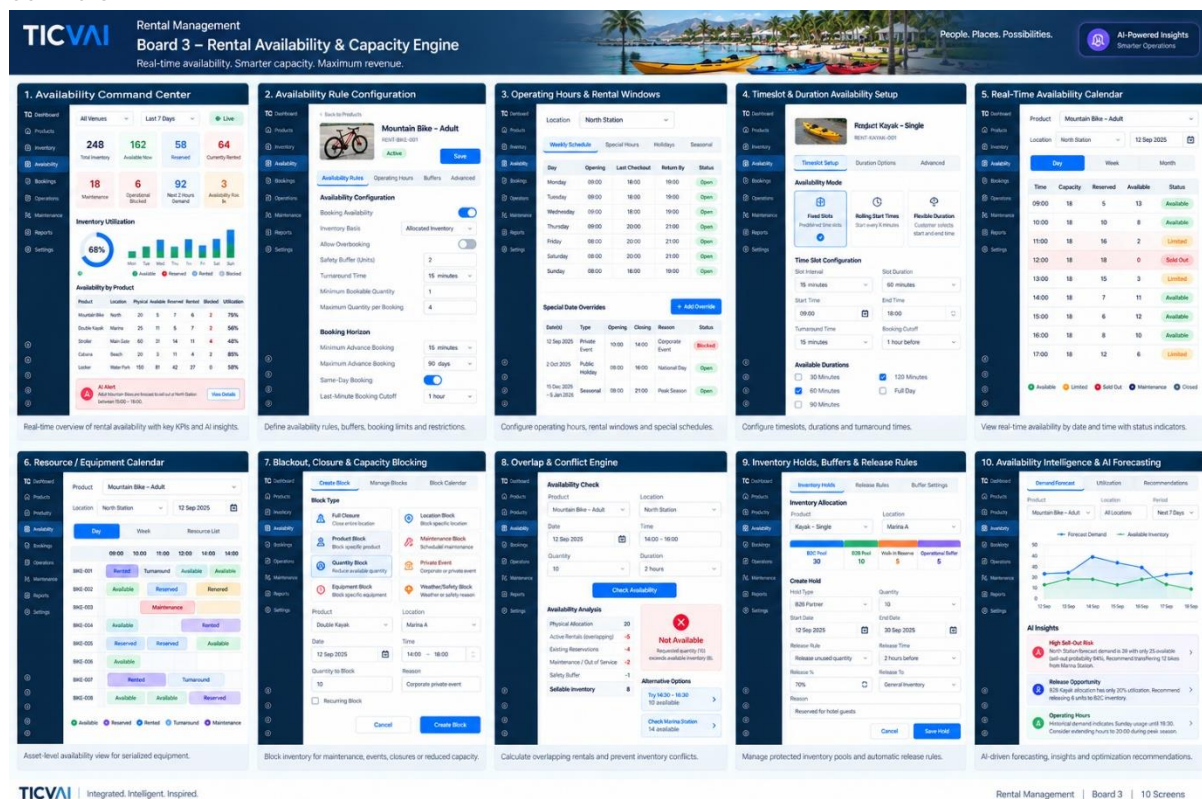
1. Manage serialized and pooled rental inventory.
2. Generate and maintain unique rental asset identities.
3. Track each serialized asset's complete lifecycle and rental history.
4. Track real-time quantities by product and location.
5. Manage operational status separately from physical condition.
6. Identify equipment using QR/barcode and support extensibility for RFID/NFC.
7. Allocate inventory across multiple rental locations.
8. Transfer serialized and pooled inventory between locations with controlled receiving.
9. Handle lost, damaged, retired and inventory-adjustment scenarios with approvals and audit trails.
10. Provide **AI-driven inventory health, shortage prediction and rebalancing recommendations**.

This gives Board 2 a very clear role: **the digital twin of TICVAI's physical rental fleet**, while Board 3 will take this inventory and determine **when and how much of it can actually be booked**.

Board 3 — Rental Availability & Capacity Engine

Module: 7 — Rental Management

Board Objective: Configure and operate the real-time engine that determines **whether a rental product can be booked, in what quantity, at which location, for what date/time, and for what duration**, while accounting for existing reservations, active rentals, maintenance, operational hours, turnaround time, buffers, and inventory conflicts.



The original requirements explicitly require real-time inventory by **product, date, time slot and location**, automatic inventory reduction/restoration, overbooking prevention, configurable buffers, operating-hour allocation, maintenance blocking and out-of-service management.

They also require the availability engine to calculate **overlapping rentals**, prevent conflicts and provide daily, weekly and resource-calendar views showing available, reserved and maintenance-blocked inventory.

Board 3 therefore answers:

“Given everything happening to our physical rental inventory, what can we safely sell right now or in the future?”

Screen 1 — Availability Command Center

Purpose

Provide management and operators with a real-time overview of rental availability across all locations.

KPI Cards

- Total Rentable Inventory
- Available Now
- Reserved
- Currently Rented
- Maintenance Blocked
- Operationally Blocked
- Next 2 Hours Demand

- Availability Risk

Availability Overview

Product	Location	Physical	Available	Reserved	Rented	Blocked	Utilization
Mountain Bike	North	20	5	7	6	2	75%
Double Kayak	Marina	25	11	5	7	2	56%
Stroller	Main Gate	60	31	14	11	4	48%
Cabana	Beach	20	3	11	4	2	85%

Filters

- Tenant
- Venue
- Rental Location
- Product
- Category
- Date
- Time
- Availability Status

Availability Health

Statuses:

Healthy / Limited / Critical / Sold Out

AI Alert

Availability Risk

Adult Mountain Bikes are forecast to sell out at North Station between 15:00–18:00.

Screen 2 — Availability Rule Configuration

Purpose

Define the fundamental rules used by the availability engine for each rental product.

Product

Mountain Bike – Adult

Availability Basis

Configure availability according to:

- Physical inventory
- Location allocation
- Existing reservations
- Active rentals
- Maintenance blocks
- Out-of-service assets
- Inventory buffer
- Turnaround time
- Operating hours

Configuration

Booking Availability: ON

Inventory Basis

Allocated Inventory

Allow Overbooking

OFF

Safety Buffer

2 Units

Turnaround Buffer

15 Minutes

Minimum Bookable Quantity

1

Maximum Quantity per Booking

4

Booking Horizon

- Minimum Advance Booking
- Maximum Advance Booking
- Same-Day Booking
- Last-Minute Booking Cutoff

Example:

Maximum advance: 90 days

Minimum advance: 15 minutes

Screen 3 — Operating Hours & Rental Windows

Purpose

Control when rental products can actually be booked and used.

The source specifically requires inventory allocation according to operating hours.

Weekly Schedule

Day	Opening	Last Checkout	Return By	Status
Monday	09:00	18:00	19:00	Open
Tuesday	09:00	18:00	19:00	Open
Wednesday	09:00	18:00	19:00	Open
Thursday	09:00	20:00	21:00	Open
Friday	08:00	20:00	21:00	Open
Saturday	08:00	20:00	21:00	Open
Sunday	08:00	18:00	19:00	Open

Important distinction:

Location Closing Time ≠ Last Rental Start Time

For example:

Location closes at **19:00**.

Maximum rental duration = **60 minutes**.

Therefore last checkout may be:

18:00

The availability engine must automatically calculate this.

Overrides

Support:

- Public holidays
 - Special events
 - Ramadan hours
 - Private events
 - Emergency closure
 - Seasonal operating hours
-

Screen 4 — Timeslot & Duration Availability Setup

Purpose

Configure how rental availability is presented to customers.

The source supports both fixed and customer-defined rental periods.

Availability Mode

Fixed Slots

Example:

Kayak Rental

09:00–10:00

10:15–11:15

11:30–12:30

Rolling Start Times

Example:

Customers may start every:

15 minutes

Duration options:

30 / 60 / 90 / 120 minutes

Flexible Duration

Customer selects:

Start: 10:30

End: 12:00

The engine checks continuous availability across the entire selected period.

Configuration

- Slot Interval
 - Fixed/Flexible Start
 - Allowed Durations
 - Minimum Duration
 - Maximum Duration
 - Turnaround Time
 - Booking Cutoff
-

Screen 5 — Real-Time Availability Calendar

Purpose

Provide the operational calendar explicitly required by the source.

Support:

Daily View | Weekly View | Product View | Location View

Example — Daily

Mountain Bike — North Station

Time	Capacity	Reserved	Available	Status
09:00	18	5	13	Available
10:00	18	10	8	Available
11:00	18	16	2	Limited
12:00	18	18	0	Sold Out
13:00	18	15	3	Limited

Color/state indicators:

- Available
- Limited
- Sold Out
- Maintenance
- Closed
- Blocked

Operators can click any period to understand **why** capacity is unavailable.

Screen 6 — Resource / Equipment Calendar

Purpose

Provide asset-level availability for serialized inventory.

This is particularly important for:

- Bikes
- Kayaks
- Watercraft
- Wheelchairs

- Cabanas
- Other individually tracked resources

Resource Timeline

Equipment	09:00	10:00	11:00	12:00	13:00
BIKE-001	Rental	Rental	Turnaround	Available	Available
BIKE-002	Available	Reserved	Reserved	Return	Available
BIKE-003	Maintenance	Maintenance	Maintenance	Maintenance	Maintenance
BIKE-004	Available	Available	Available	Reserved	Reserved

Selecting a block shows:

- Reservation
- Customer
- Expected checkout
- Expected return
- Turnaround
- Equipment status
- Related maintenance

The source specifically requests a **Resource Calendar View** alongside daily and weekly availability.

Screen 7 — Blackout, Closure & Capacity Blocking

Purpose

Allow authorized users to deliberately remove inventory from sale.

The original scope requires seasonal availability, blackout dates and maintenance periods.

Create Block

Block Type

- Full Closure
- Product Block
- Location Block
- Quantity Block
- Equipment Block
- Maintenance Block
- Private Event
- Operational Block
- Weather/Safety Block

Example

Product: Double Kayak

Location: Marina A

Date: 12 September

Time: 14:00–18:00

Block: 10 Units

Reason:

Corporate private event.

Recurring Blocks

Support:

- Every Monday
- Selected weekdays
- Date range
- Seasonal period

System must immediately recalculate sellable availability.

Screen 8 — Overlap & Conflict Engine

Purpose

Provide transparency into one of the most important calculations in the rental system.

The original requirement explicitly states:

Calculate overlapping rentals and prevent inventory conflicts.

Example:

Customer requests:

10 Mountain Bikes

14:00–16:00

Engine evaluates:

Physical Allocation: 20

Less:

- Active rentals overlapping period: 5
- Existing reservations: 4
- Maintenance: 2
- Safety buffer: 1

Sellable inventory:

8

Requested:

10

Result:

✗ NOT AVAILABLE

Alternative recommendations:

14:30 — 10 available

or

Marina Station — 14 available

Conflict Types

- Reservation overlap
- Rental extension overlap
- Maintenance conflict
- Turnaround conflict
- Location conflict
- Inventory shortage

This engine should be used consistently by B2C, B2B, POS and staff booking channels.

Screen 9 — Inventory Holds, Buffers & Release Rules

Purpose

Control inventory that should temporarily or permanently be withheld from normal sales.

The source explicitly requires configurable inventory buffers.

Buffer Types

Safety Buffer

Example:

Physical inventory: 50

Safety buffer: 5

Maximum sellable:

45

Operational Hold

Temporarily reserve inventory for:

- VIP
- Hotel guests
- B2B partner
- Walk-in customers
- Event operations
- Emergency replacement
- Internal use

Example:

50 Kayaks

B2C Pool: 30

B2B Pool: 10

Walk-In Reserve: 5

Operational Buffer: 5

Release Rules

This is one of my recommended enhancements.

Example:

If B2B allocation remains unsold **2 hours before the rental period**, release 70% to general inventory.

Configure:

Hold → Release Time → Release Quantity/% → Destination Pool

This prevents protected inventory from remaining unused unnecessarily.

Screen 10 — Availability Intelligence & AI Forecasting

Purpose

Use AI to forecast demand and proactively optimize rental availability.

The source recommends AI-based utilization and demand forecasting.

Forecast Dashboard

Display:

- Forecast Demand
- Forecast Utilization
- Sell-Out Probability
- Inventory Shortage Risk
- Excess Inventory
- Location Imbalance
- Expected Cancellations
- Expected Late Returns

Example AI Insight

Saturday — Mountain Bikes

North Station

Available inventory: **25**

Forecast demand: **39**

Sell-out probability: **94%**

Recommended action:

Transfer 12 bikes from Marina Station before 13:00.

AI Release Recommendation

B2B Kayak allocation has only 20% utilization with 90 minutes remaining before release cutoff.

Recommend releasing **6 units** to B2C inventory.

AI Operating Recommendation

Historical demand indicates Sunday rental demand continues until 19:30.

Consider extending rental operating hours from 19:00 to 20:00 during peak season.

AI Turnaround Recommendation

Average kayak inspection/cleaning takes 8 minutes. Current configured turnaround is 20 minutes.

Reducing turnaround to 15 minutes could increase daily capacity by approximately 7%, subject to operational approval.

All AI recommendations should remain **advisory**, with authorized users approving changes.

Important Availability Formula

I recommend explicitly documenting the fundamental logic for Softlabs:

Sellable Availability

Allocated Physical Inventory

– Active Rentals Overlapping Requested Period

– Existing Reservations

– Maintenance / Out-of-Service Inventory

– Operational Blocks

– Safety Buffers / Protected Inventory

+ Eligible Released Inventory

= Sellable Availability

And this calculation must apply across the **entire requested rental duration + turnaround period**, not simply at the starting time.

That is extremely important.

For example, if a bike is available at **10:00** but already reserved at **11:00**, TICVAI must not sell it for a **10:00–12:00** rental.

Board 3 — Core Flow

Physical Inventory

↓

Location Allocation

↓

Operating Hours

↓

Duration + Turnaround

↓

Existing Reservations / Active Rentals

↓

Maintenance & Blocks

↓

Buffers / Protected Inventory

↓

REAL-TIME AVAILABILITY ENGINE

↓

B2C | B2B | POS | Operator | API

This should become the **single availability engine** for all TICVAI rental sales channels.

Integration Boundaries

To prevent duplication:

Board 2 owns:

Physical quantity, serial numbers, equipment status and location allocation.

Board 3 owns:

Calculation of **when and how much inventory is bookable**.

Board 4 owns:

What price and deposit apply to the available rental.

Board 5 owns:

Creating the actual reservation.

Board 7 owns:

Extensions and active rental operations—but any extension must call Board 3's availability engine before approval.

Board 9 owns:

Maintenance scheduling, while Board 3 consumes maintenance blocks to remove affected equipment from availability.

Board 3 — Acceptance Criteria

Board 3 is complete when TICVAI can:

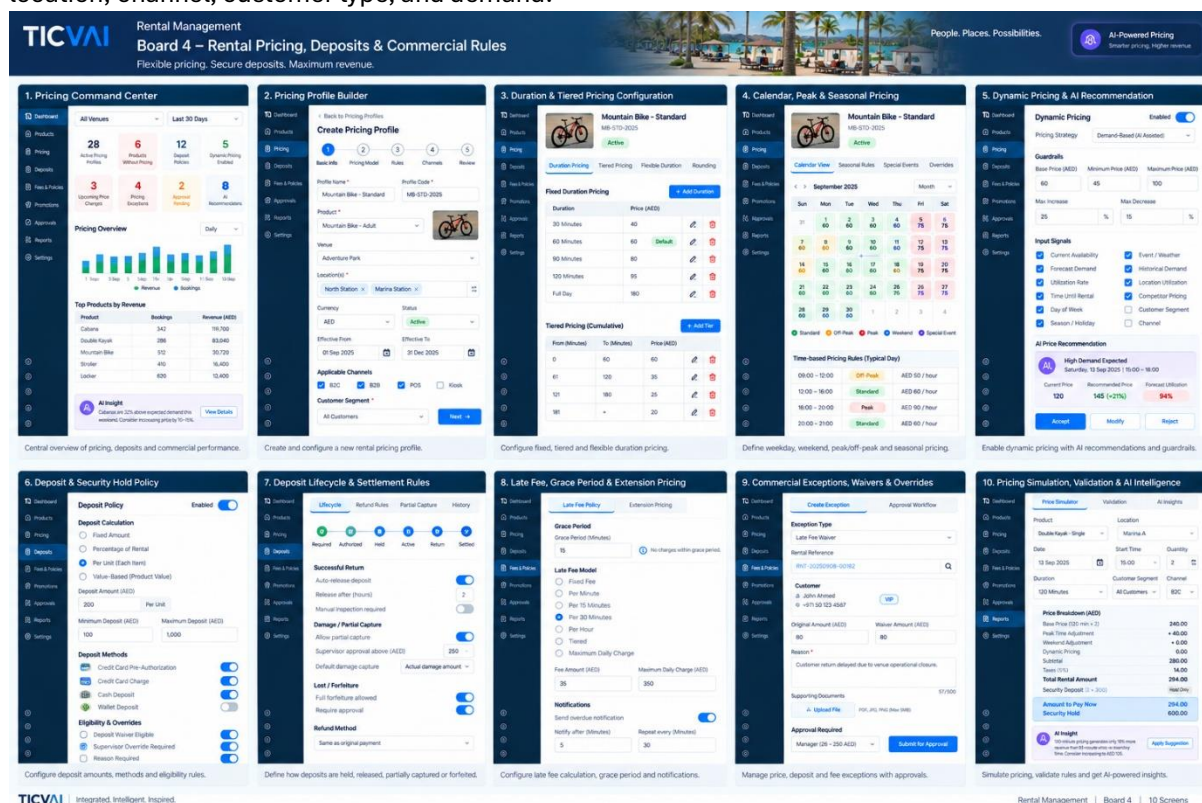
1. Calculate real-time rental availability by **product, date, time, duration and location**.
2. Prevent overbooking and overlapping rental conflicts.
3. Support fixed slots, rolling starts and flexible rental periods.
4. Apply operating hours and automatically determine valid last checkout times.
5. Account for turnaround/cleaning periods.
6. Provide daily, weekly and serialized-resource calendar views.
7. Apply blackout dates, maintenance blocks, closures and operational capacity reductions.
8. Support inventory safety buffers and protected inventory pools.
9. Automatically release unused protected inventory according to configurable rules.
10. Feed one consistent availability result to **B2C, B2B, POS, operators and APIs**.
11. Recalculate availability immediately after reservation, cancellation, return, maintenance or inventory-status changes.
12. Use AI to forecast shortages, sell-outs, utilization and inventory imbalances and recommend corrective actions.

This Board 3 becomes the **central brain controlling rental capacity**. It ensures TICVAI never looks only at “how many bikes we own,” but understands **how many can actually be sold for this exact location, date, start time and rental duration**.

Board 4 — Rental Pricing, Deposits & Commercial Rules

Module: 7 — Rental Management

Board Objective: Configure how every rental is priced, how deposits/security holds are calculated and collected, how extensions and late returns are charged, and which commercial rules apply based on duration, date, time, location, channel, customer type, and demand.



The original requirements support **fixed and customer-defined rental durations**, together with flat-rate, tiered, dynamic, peak/off-peak, weekend, and seasonal pricing.

They also require fixed/percentage deposits, credit-card pre-authorization, cash/wallet deposits, full or partial refunds, forfeiture, automatic release after successful return, and complete deposit transaction history.

The source additionally requires **automatic late-fee calculation, configurable grace periods, overdue notifications and manager approval for fee waivers**.

Board 4 answers:

“How much should this rental cost, what financial security do we require, and what happens commercially when the rental changes or something goes wrong?”

I recommend exactly **10 screens**.

Screen 1 — Rental Pricing Command Center

Purpose

Central management screen for all rental pricing and commercial policies.

KPI Cards

- Active Pricing Profiles
- Products Without Pricing
- Deposit Policies
- Dynamic Pricing Enabled
- Upcoming Price Changes

- Pricing Exceptions
- Approval Pending
- AI Recommendations

Pricing Directory

Product	Location	Base Price	Model	Deposit	Dynamic	Status
Mountain Bike	North	AED 60	Tiered	AED 200	✓	Active
Double Kayak	Marina	AED 120	Peak/Off-Peak	AED 300	✓	Active
Stroller	Main Gate	AED 40	Flat	AED 100	—	Active
Cabana	Beach	AED 350	Seasonal	25%	✓	Active

Filters

- Venue
- Location
- Product
- Category
- Pricing Model
- Deposit Type
- Effective Date
- Status

Actions

+ Create Pricing Profile

Additional:

Edit | Duplicate | Simulate | Submit for Approval | Schedule

Screen 2 — Pricing Profile Builder

Purpose

Create the master commercial pricing profile associated with a rental product.

Basic Information

- Profile Name
- Profile Code
- Product
- Venue
- Location(s)
- Currency
- Sales Channel
- Customer Segment
- Effective From
- Effective To
- Status

Pricing Model

Select:

- Flat Rate
- Duration Based
- Tiered
- Peak / Off-Peak
- Weekend
- Seasonal
- Dynamic / AI-Assisted
- Hybrid

Example

Mountain Bike – Standard Pricing

Currency: AED

Base Duration: 60 Minutes

Base Price: 60

Additional Duration:

AED 20 / 30 minutes

Priority

If several pricing profiles apply, TICVAI must determine the applicable profile using configurable precedence.

Screen 3 — Duration & Tiered Pricing Configuration

Purpose

Configure pricing according to rental duration.

This directly implements the fixed and dynamic-duration requirements.

Fixed Duration Pricing

Duration	Price
30 Minutes	AED 40
60 Minutes	AED 60
90 Minutes	AED 80
120 Minutes	AED 95
Full Day	AED 180

Tiered Pricing

Example:

First Hour: AED 60

Second Hour: +AED 35

Third Hour: +AED 25

Additional Hour: +AED 20

This allows the effective hourly rate to decrease for longer rentals.

Flexible Duration

For customer-defined durations:

Minimum Charge: AED 40

Billing Increment: 15 minutes

Additional Increment: AED 10

Rounding Rules

Configure:

- Exact usage
 - Round up to 15 minutes
 - Round up to 30 minutes
 - Full next hour
-

Screen 4 — Calendar, Peak & Seasonal Pricing

Purpose

Allow price differentiation according to date and time.

Pricing Calendar

Configure:

Weekday

AED 60/hour

Weekend

AED 75/hour

Peak

16:00–20:00 → AED 90/hour

Off-Peak

09:00–12:00 → AED 50/hour

Seasonal Pricing

Examples:

Peak Season

1 Nov – 31 Mar

Summer

1 Jun – 31 Aug

Public Holiday

Special pricing

Special Event

Event-specific pricing

Calendar View

Use visual indicators to show:

Standard | Peak | Off-Peak | Weekend | Seasonal | Special Event

Conflict Resolution

If multiple pricing rules overlap, system shall apply configured priority.

For example:

Special Event > Holiday > Seasonal > Weekend > Standard

Screen 5 — Dynamic Pricing & AI Recommendation

Purpose

Connect Rental Management to TICVAI's broader Dynamic Pricing capability without duplicating the core Dynamic Pricing module.

The original rental requirements explicitly support dynamic pricing.

Dynamic Pricing

Enabled: ON

Input Signals

Potential signals:

- Current availability
- Forecast demand
- Utilization
- Time until rental
- Day of week
- Season
- Weather/event signal where integrations exist
- Historical demand
- Location utilization

Guardrails

Base Price:

AED 60

Minimum:

AED 45

Maximum:

AED 100

Maximum increase:

+25%

Maximum decrease:

-15%

AI Recommendation

Double Kayak – Saturday 15:00–18:00

Forecast utilization: **94%**

Current Price: **AED 120**

Recommended Price: **AED 145**

Expected Revenue Impact: **+11%**

Actions:

Accept | Modify | Reject | Schedule

AI must never exceed configured commercial guardrails.

Screen 6 — Deposit & Security Hold Policy

Purpose

Configure financial security required before equipment is released.

The original scope supports fixed and percentage deposits and multiple deposit methods.

Deposit Requirement

Deposit Required: YES

Deposit Calculation

Select:

Fixed Amount

AED 300

or

Percentage

30%

I also recommend:

Per Unit

Example:

AED 200 × each bicycle

and:

Risk/Value-Based Deposit

for higher-value equipment, subject to client configuration.

Deposit Methods

Enable:

- Credit Card Pre-Authorization
- Card Charge
- Cash
- Wallet

Example

Rental:

3 Mountain Bikes

Deposit Rule:

AED 200/unit

Required Deposit:

AED 600

Limits

- Minimum Deposit
- Maximum Deposit
- Deposit Currency
- Deposit Waiver Eligibility
- Supervisor Override

Screen 7 — Deposit Lifecycle & Settlement Rules

Purpose

Control what happens to the deposit throughout the rental lifecycle.

Lifecycle

Required

- **Authorized/Collected**
- **Held**
- **Rental Active**
- **Return Inspection**

Then:

Successful Return

Release Full Deposit

Damage

Partial Capture / Partial Refund

Lost Equipment

Full or Configured Forfeiture

Deposit Rules

Configure:

- Automatic release
- Manual inspection required
- Supervisor approval threshold
- Partial capture permitted
- Full capture permitted
- Auto-release delay
- Refund method

Example

Deposit:

AED 500

Damage Charge:

AED 120

Result:

Capture: AED 120

Release: AED 380

Actual payment execution must integrate with TICVAI's Payment/Finance layer rather than create a separate payment engine.

Screen 8 — Late Fee, Grace Period & Extension Pricing

Purpose

Configure the commercial treatment of rentals that extend beyond the original return time.

The source explicitly requires automatic late-fee calculation and grace-period configuration.

Grace Period

15 Minutes

Late Fee Model

Select:

- Fixed Fee
- Per Minute
- Per 15 Minutes
- Per 30 Minutes
- Per Hour
- Tiered
- Maximum Daily Charge

Example

Expected Return:

14:00

Actual Return:

15:12

Grace:

15 min

Chargeable Late Time:

57 min

Late Fee:

AED 40

Extension Pricing

I recommend separating:

Customer-approved extension

from:

Unauthorized late return

Example:

Normal extension:

AED 20 / 30 min

Late return:

AED 35 / 30 min

This encourages customers to extend properly rather than simply keep the equipment.

Extension availability itself must be validated by Board 3 before extension is approved.

Screen 9 — Commercial Exceptions, Waivers & Overrides

Purpose

Control authorized deviations from normal commercial policies.

Exception Types

- Price Override

- Complimentary Rental
- Deposit Waiver
- Deposit Reduction
- Late Fee Waiver
- Damage Fee Waiver
- Extension Fee Waiver
- Manual Refund
- Goodwill Adjustment

Example

Rental: RNT-20260908-00182

Late Fee:

AED 80

Requested Waiver:

AED 80

Reason:

Customer return delayed due to venue operational closure.

Approval

Cashier → Supervisor Approval Required

The original requirement explicitly requires manager approval for late-fee waivers.

Approval Thresholds

Example:

Up to AED 25 → Supervisor

AED 26–250 → Manager

Above AED 250 → Administrator

Every override must record:

- Original Amount
- Adjusted Amount
- Reason
- Requested By
- Approved By
- Date/Time

Screen 10 — Pricing Simulation, Validation & AI Commercial Intelligence

Purpose

Allow administrators to test commercial configuration before publishing it.

This is particularly important because rental pricing can involve duration, calendar, location, deposit and dynamic pricing simultaneously.

Price Simulator

Enter:

Product: Double Kayak

Location: Marina A

Date: Saturday, 12 September

Start: 16:00

Duration: 120 Minutes

Quantity: 2
Channel: B2C

Calculation

Base Rental:

$$2 \times \text{AED } 120 = \text{AED } 240$$

Peak Adjustment:

$$+ \text{AED } 40$$

Weekend Adjustment:

Included in profile

Rental Total:

AED 280

Deposit:

$$2 \times \text{AED } 300$$

AED 600 Hold

Customer Pays Now:

AED 280

Security Hold:

AED 600

Explain Price

The system should show **why the price was calculated**.

This is important for customer service and auditability.

AI Commercial Insight

Current 120-minute Mountain Bike pricing generates only 18% more revenue than the 60-minute product while consuming twice the inventory duration.

Consider increasing the 120-minute rate from AED 95 to AED 105.

Validation

Check for:

- Missing prices
- Conflicting rules
- Invalid effective dates
- Missing deposit policy
- Dynamic price outside guardrails
- Pricing gaps
- Channel conflicts

Actions:

Save Draft | Simulate | Submit for Approval | Publish

Important Commercial Calculation Logic

I recommend that Softlabs make the calculation transparent rather than simply returning one number.

Rental Charge

Base/Duration Price

± Calendar/Season Rule

± Applicable Dynamic Pricing

– Promotion/Discount

+ Approved Add-ons

+ Applicable Taxes/Fees

= Rental Amount

Then separately:

Security Deposit

Deposit Policy × Quantity/Value

= Deposit / Authorization Amount

The deposit should **not be treated as rental revenue** unless an amount is legitimately captured or forfeited.

Important Extension Logic

When the customer asks:

“Can I keep the bike another hour?”

TICVAI should not simply charge another hour.

The workflow must be:

Extension Request

→ **Board 3 Availability Check**

→ Is equipment available for requested extension?

If **YES**:

Calculate Extension Price → Customer Accepts → Payment → Extend Rental

If **NO**:

Extension cannot be accepted because the equipment is required for another reservation at 16:00.

This integration between Board 3 and Board 4 is extremely important.

Pricing Rule Priority

We should also specify a configurable hierarchy so Softlabs doesn't create unpredictable pricing.

Example:

Product Base Price

↓

Location Rule

↓

Date/Season Rule

↓

Dynamic Pricing



Customer/Channel Eligibility



Promotion



Authorized Manual Override



FINAL RENTAL PRICE

The exact precedence must be configurable and auditable.

Board 4 — Core Flow

Select Rental Product



Create Pricing Profile



Configure Duration Pricing



Configure Calendar / Peak Pricing



Configure Dynamic Pricing



Configure Deposit



Configure Late & Extension Fees



Configure Approval Rules



Simulate



Approve & Publish

Integration Boundaries

To avoid duplication:

Board 4 owns:

Rental-specific pricing profiles, duration pricing, rental deposit policy, extension fees, late-fee policy and commercial exceptions.

Dynamic Pricing Module owns:

The central AI/dynamic-pricing engine, models, experimentation, enterprise guardrails and broader dynamic-pricing governance.

Promotion Module owns:

Coupons, promotions, campaigns and promotional eligibility.

Payments owns:

Authorization, capture, refund and payment-provider processing.

Finance owns:

Accounting, reconciliation, settlement and financial posting.

Board 3 owns:

Availability. Pricing must never create inventory.

Board 7 owns:

The operational extension request, while Board 4 calculates its price.

Board 8 owns:

Final return/damage settlement, while Board 4 provides applicable fee/deposit policies.

Board 4 — Acceptance Criteria

Board 4 is complete when TICVAI can:

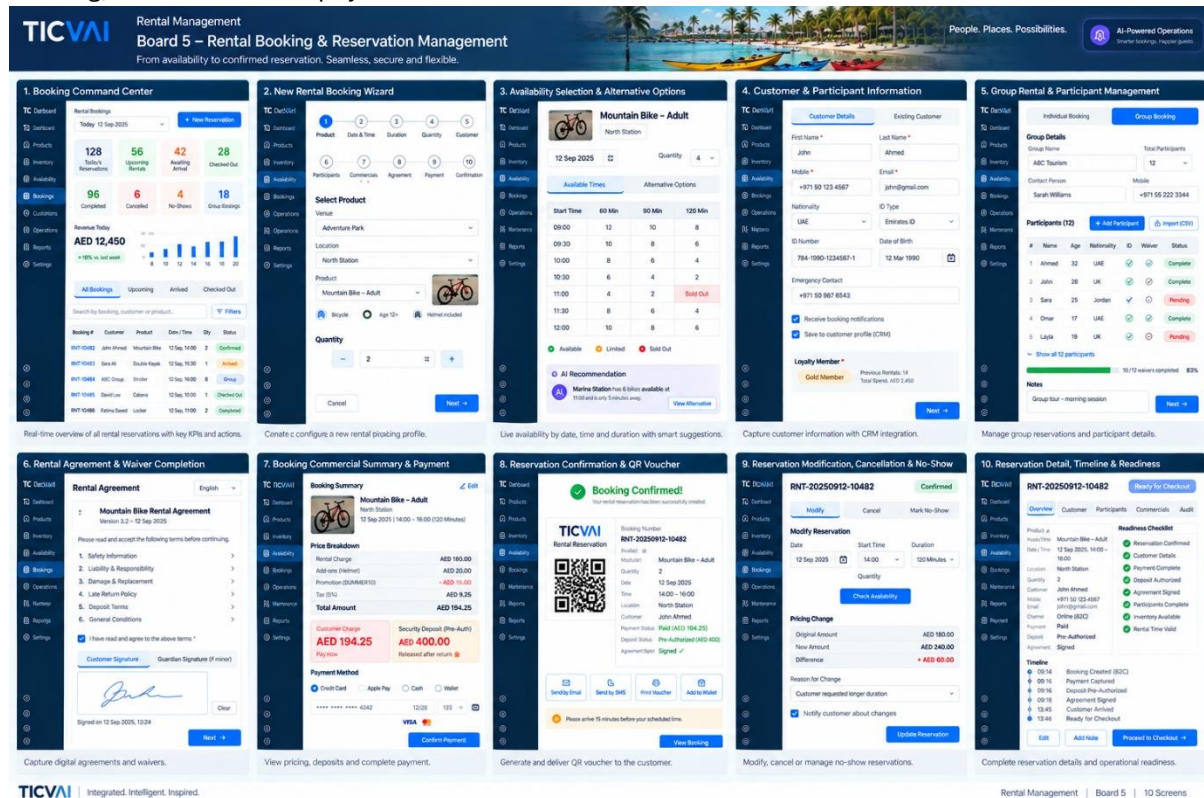
1. Configure rental pricing by **product, duration, location, date, time and channel**.
2. Support flat, duration, tiered, peak/off-peak, weekend, seasonal and dynamic pricing.
3. Support fixed and customer-defined rental durations.
4. Configure AI/dynamic pricing with strict minimum/maximum guardrails.
5. Configure fixed, percentage and per-unit deposits.
6. Support card pre-authorization, cash and wallet deposit policies.
7. Manage full release, partial capture/refund and deposit forfeiture rules.
8. Configure extension pricing independently from unauthorized late-return fees.
9. Automatically calculate late fees with configurable grace periods.
10. Support controlled price/deposit/fee overrides with approval workflows.
11. Simulate and explain the complete rental price before publication.
12. Maintain effective dates, versions, approvals and full commercial audit history.
13. Integrate cleanly with **Availability, Dynamic Pricing, Promotions, Payments and Finance** without duplicating those engines.
14. Provide **AI recommendations for pricing optimization and commercial performance**.

This makes Board 4 the **commercial brain of Rental Management**: Board 3 determines **whether we can sell the rental**, while Board 4 determines **what we should charge and what financial protection is required**.

Board 5 — Rental Booking & Reservation Management

Module: 7 — Rental Management

Board Objective: Manage the complete rental reservation lifecycle from availability selection through customer/participant capture, reservation creation, voucher generation, modification, cancellation, group booking, and readiness for physical checkout.



The original requirements define two primary rental booking models:

- **Voucher-Based Rental:** customer selects the rental item, quantity and duration, provides customer information, and receives a voucher/ticket that can later be redeemed at the rental counter.
- **Time-Slot Rental:** customer selects product, date, time, duration and quantity, with the system validating overlapping rentals and real-time availability.

The requirements also support configurable customer fields, participant demographics and group reservations.

Board 5 answers:

“How does an available rental become a confirmed customer reservation?”

I recommend exactly **10 screens**.

Screen 1 — Rental Booking Command Center

Purpose

Provide a single operational view of all rental reservations across venues and locations.

KPI Cards

- Today's Reservations
- Upcoming Rentals
- Awaiting Arrival
- Checked Out
- Completed

- Cancelled
- No-Shows
- Group Bookings
- Revenue Today
- Booking Alerts

Reservation Table

Booking	Customer	Product	Date/Time	Qty	Location	Status
RNT-10482	John Ahmed	Mountain Bike	14:00	2	North	Confirmed
RNT-10483	Sara Ali	Double Kayak	15:30	1	Marina	Arrived
RNT-10484	ABC Group	Stroller	16:00	8	Main Gate	Group

Reservation Status

- Draft
- Pending Payment
- Confirmed
- Arrived
- Ready for Checkout
- Checked Out
- Partially Returned
- Completed
- Cancelled
- No-Show

Filters

- Date
- Venue
- Location
- Product
- Booking Type
- Channel
- Customer
- Status

Actions:

+ New Reservation | Scan Voucher | Find Booking

Screen 2 — New Rental Booking Wizard

Purpose

Allow POS/rental operators to create a reservation through a guided workflow.

Booking Steps

1. Product

- 2. Date & Time
- 3. Duration
- 4. Quantity

- 5. Customer
- 6. Participants
- 7. Commercials
- 8. Agreement
- 9. Payment
- 10. Confirmation

Step 1

Select:

- Venue
- Rental Location
- Rental Product
- Quantity

Step 2

Select:

- Rental Date
- Start Time
- Duration

Board 5 must call **Board 3 Availability Engine** immediately.

Example:

- ✓ 12 Mountain Bikes available
Saturday, 14:00–16:00

Commercial Call

Board 4 returns:

- Rental Price:** AED 120
- Deposit:** AED 400

The booking wizard itself should not recalculate pricing or availability independently.

Screen 3 — Availability Selection & Alternative Options

Purpose

Give the operator/customer a clear booking choice based on Board 3 availability.

Example

Mountain Bike — Adult

12 September 2026

Start	60 Min	90 Min	120 Min
13:00	12	10	8
13:30	10	8	6
14:00	8	6	Sold Out
14:30	12	10	7
15:00	14	11	9

If requested inventory is unavailable:

Requested

4 Kayaks
14:00–16:00

✕ Only 2 Available

TICVAI should suggest:

14:30 — 4 available

15:00 — 7 available

or:

Marina B — 6 available at 14:00

AI Recommendation

Marina B has immediate availability and is approximately 5 minutes from the requested rental location.

The final decision remains with the customer/operator.

Screen 4 — Customer & Participant Information

Purpose

Capture the configurable customer information required by the rental product.

The original requirements specifically support configurable mandatory fields such as name, mobile, email, nationality, ID number and date of birth.

Primary Renter

- First Name
- Last Name
- Mobile
- Email
- Nationality
- ID Type
- ID Number
- Date of Birth
- Emergency Contact

Fields shall follow the product configuration created in **Board 1**:

Required | Optional | Hidden

Existing Customer

Search:

Mobile | Email | Customer ID | Membership

If customer exists in CRM:

John Ahmed

Previous Rentals: 14
Loyalty Member: Gold

Avoid creating duplicate customer records.

Privacy

Customer information must follow TICVAI's broader consent/privacy policies.

Screen 5 — Group Rental & Participant Management

Purpose

Handle reservations involving multiple participants.

The source explicitly requires group reservations, participant demographics and configurable participant forms.

Group Booking

Booking: RNT-GRP-0092

Primary Booker:

ABC Tourism

Participants:

12

Equipment:

12 Mountain Bikes

Participant Table

Participant	Age	Nationality	ID	Waiver	Equipment
Ahmed	32	UAE	✓	Signed	Pending
John	28	UK	✓	Signed	Pending
Sara	25	Jordan	✓	Pending	Pending

Support

- Add participant
- Bulk upload participants
- Copy primary contact
- Participant-specific fields
- Individual waivers
- Guardian information
- Group waiver where allowed

Group Readiness

10/12 Waivers Complete

Alert:

⚠ 2 participants cannot proceed to checkout until required waivers are completed.

Screen 6 — Rental Agreement & Waiver Completion

Purpose

Complete all mandatory customer acknowledgments before confirmation or checkout.

This implements the digital rental agreement/waiver capability recommended in the source.

Agreement

Mountain Bike Rental Agreement — Version 3.2

Display:

- Rental Terms
- Liability Terms
- Safety Conditions
- Damage Responsibility
- Late Return Policy
- Deposit Policy
- Cancellation Policy

Completion Methods

- Sign on POS/tablet
- Customer mobile link
- B2C online signing
- QR self-service signing

E-Signature

Customer Signature

Date/Time automatically captured.

For minors:

Guardian Signature Required

Status

- ✓ Rental Agreement
- ✓ Safety Declaration
- ✓ Deposit Terms
- ⚠ Participant 4 Waiver

Signed agreement must be attached to the reservation record with the applicable version.

Screen 7 — Booking Commercial Summary & Payment

Purpose

Show the complete financial commitment before confirming the reservation.

Rental Summary

Mountain Bike – Adult × 2

Date: 12 Sep
Time: 14:00
Duration: 120 min

Charges

Rental:

AED 190

Add-ons:

AED 20

Promotion:

– AED 15

Tax/Fees:

AED 10

Amount Payable

AED 205

Security Deposit

AED 400 Pre-Authorization

Clearly separate:

CUSTOMER CHARGE

AED 205

from:

SECURITY HOLD

AED 400

Payment Status

- Paid
- Partially Paid
- Pay at Counter
- Pre-Authorized
- Pending

Payment execution must use TICVAI's Payments module.

Screen 8 — Reservation Confirmation & QR Voucher

Purpose

Create the confirmed rental reservation and customer credential.

The original voucher flow specifically requires the system to generate a voucher/ticket that is later redeemed at the rental counter.

Confirmation

BOOKING CONFIRMED

RNT-20260912-10482

Mountain Bike × 2

12 September

14:00–16:00

North Station

QR Voucher

[QR CODE]

Voucher Contents

- Reservation ID
- Product

- Quantity
- Date
- Start time
- Duration
- Location
- Customer
- Payment status
- Deposit status
- Agreement status

Delivery

- Email
- SMS
- Mobile App
- Wallet
- Print

Actions

View Booking | Send Again | Print Voucher | Add to Wallet

Screen 9 — Reservation Modification, Cancellation & No-Show

Purpose

Manage the reservation after confirmation but before/during fulfillment.

Modify Reservation

Allow authorized changes to:

- Date
- Time
- Duration
- Quantity
- Product
- Rental Location
- Participants

Critical Rule

Every modification affecting inventory must trigger:

Board 3 Availability Revalidation

Every modification affecting price must trigger:

Board 4 Price Recalculation

Example:

Original:

14:00–15:00
AED 60

Requested:

14:00–16:00

Availability:

✓ Available

New Price:

AED 95

Difference:

+ AED 35

Customer confirms before modification.

Cancellation

Display:

- Cancellation policy
- Refund amount
- Deposit release
- Cancellation fee
- Inventory released

Once cancelled, reserved inventory must automatically return to availability as required by the source.

No-Show

This is one of our recommended enhancements.

Configure:

Expected arrival:

14:00

Grace:

30 Minutes

At 14:30:

Mark No-Show / Extend Hold

Based on policy, TICVAI may:

- Release inventory
- Apply no-show charge
- Release/capture deposit as applicable
- Notify customer

Screen 10 — Reservation Detail, Timeline & Readiness

Purpose

Provide the complete operational record before handing the reservation to Board 6 for checkout.

Reservation Header

RNT-20260912-10482

Status:

READY FOR CHECKOUT

Tabs

Overview | Customer | Participants | Commercials | Agreement | Payments | Changes | Audit

Readiness Checklist

- ✓ Reservation Confirmed
- ✓ Customer Details
- ✓ Payment Complete
- ✓ Deposit Authorized
- ✓ Agreement Signed
- ✓ Participants Complete
- ✓ Inventory Available
- ✓ Rental Time Valid

Checkout Readiness

8 / 8 Requirements Complete

Next Action

Proceed to Checkout

This transfers the operational flow into **Board 6 — Rental Checkout & Equipment Assignment**.

Timeline

Example:

09:14 Booking Created — B2C

09:16 Payment Captured

09:16 Deposit Pre-Authorized

09:18 Agreement Signed

13:52 Customer Arrived

13:53 Ready for Checkout

Every reservation action should be auditable.

Important Booking Architecture

I strongly recommend that Softlabs does **not build separate booking logic for each channel**.

There should be one central:

RENTAL RESERVATION ENGINE

used by:

B2C

B2B

POS

Rental Operator

Mobile App

Kiosk where applicable

API / Third Party

The user interfaces may differ, but availability, pricing, reservation and commercial rules must come from the same backend services.

This avoids a situation where B2C says:

4 bikes available

while POS says:

2 bikes available.

Voucher Booking vs Time-Slot Booking

The source defines both flows, and I recommend preserving both because they serve different businesses.

Flow A — Voucher Based

Product → Quantity → Duration → Customer → Payment → Voucher

The customer later arrives and the actual equipment is assigned.

Useful for:

- Strollers
- Wheelchairs
- General bicycle rental
- Some lockers

Flow B — Time-Slot Based

Product → Date → Time → Duration → Quantity → Availability → Payment → Reservation

Useful for:

- Kayaks
- Watercraft
- Cabanas
- High-demand equipment
- Limited-capacity rentals

The rental product configuration should determine which flow applies.

Reservation vs Equipment Assignment

Another important architectural distinction:

At booking time:

Customer books **2 Mountain Bikes**

We should normally **not assign BIKE-001 and BIKE-002 yet.**

The reservation holds:

Product Quantity = 2

When the customer physically arrives:

Scan Reservation

↓

Select/Scan BIKE-017

Select/Scan BIKE-028

↓

Assign to Customer

That belongs to **Board 6.**

This provides much greater operational flexibility because equipment can become damaged, require maintenance, or move location between booking and arrival.

For certain resources such as a specific **Cabana #12** or **Locker #304**, however, TICVAI should support optional **resource-at-booking assignment.**

Board 5 — Core Booking Flow

Select Product



Check Availability — Board 3



Calculate Price & Deposit — Board 4



Capture Customer



Capture Participants



Complete Agreement/Waiver



Payment / Deposit Authorization



Create Reservation



Generate QR Voucher



Customer Arrives



BOARD 6 — CHECKOUT

Integration Boundaries

Board 5 owns

- Reservation creation
- Customer/participant association
- Group booking
- Reservation lifecycle
- Voucher generation
- Booking modification
- Cancellation
- No-show
- Booking readiness

Board 3 owns

Availability and capacity calculations.

Board 4 owns

Rental pricing, deposits, extension and fee rules.

CRM owns

Customer master profile and broader customer history/segmentation.

Payments owns

Actual payment authorization, capture and refund.

Ticket Media / Credential layer owns

Reusable QR/media technology where appropriate.

Board 6 owns

Physical equipment assignment and checkout.

Board 5 — Acceptance Criteria

Board 5 is complete when TICVAI can:

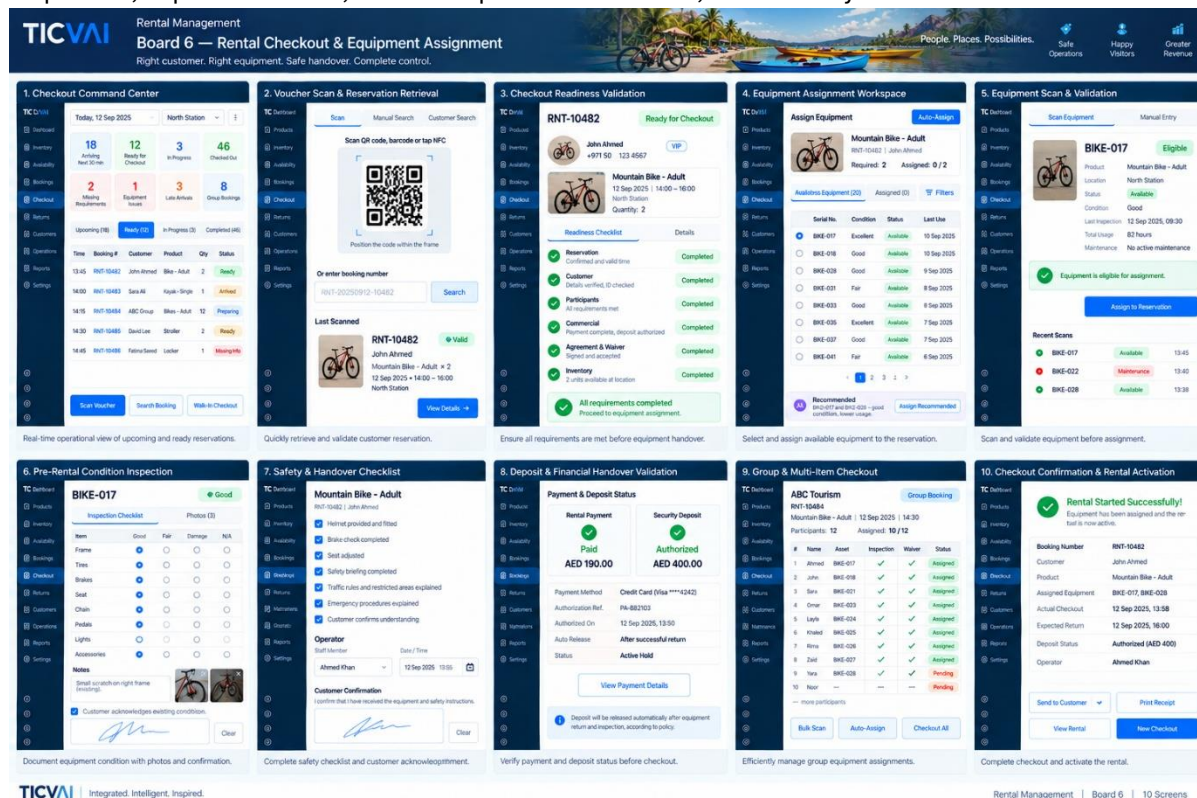
1. Create **voucher-based and time-slot rental reservations**.
2. Use one centralized reservation engine across B2C, B2B, POS, operator and API channels.
3. Validate real-time availability through Board 3 before confirming inventory.
4. Retrieve pricing and deposit requirements from Board 4.
5. Capture configurable customer information.
6. Support existing-customer lookup and CRM linkage.
7. Manage group reservations and individual participants.
8. Capture digital rental agreements, waivers and guardian consent where required.
9. Process the booking's payment/deposit workflow through TICVAI Payments.
10. Generate a unique QR rental voucher.
11. Support booking modifications with automatic availability and price recalculation.
12. Support cancellations with automatic inventory restoration.
13. Support configurable no-show/grace-period handling and inventory release.
14. Maintain complete booking history and audit trail.
15. Validate all prerequisites and provide a clear **Ready for Checkout** status.
16. Pass the confirmed reservation cleanly to **Board 6 for physical equipment assignment and checkout**.

This makes Board 5 the point where **availability + pricing + customer + agreement + payment become an actual rental reservation**.

Board 6 — Rental Checkout & Equipment Assignment

Module: 7 — Rental Management

Board Objective: Manage the physical handover of rental equipment to the customer: retrieve and validate the reservation, confirm operational readiness, assign the actual serialized or pooled equipment, perform pre-rental inspection, capture evidence, confirm deposit/waiver status, and officially start the rental.



The original requirement states that when the customer arrives, TICVAI must support **scanning the voucher/ticket, validating the booking, assigning the actual rental item, scanning the equipment serial number, linking the equipment to the reservation, and recording checkout time, staff member, assigned serial number and deposit status.**

Board 6 answers:

“The customer has arrived—what exact equipment are we giving them, is everything ready and safe, and when does the rental officially start?”

I recommend exactly **10 screens**.

Screen 1 — Rental Checkout Command Center

Purpose

Provide rental operators with a real-time operational queue of customers requiring equipment checkout.

KPI Cards

- Arriving Next 30 Min
- Awaiting Checkout
- Ready for Checkout
- Checkout in Progress
- Checked Out Today
- Missing Requirements
- Equipment Issues

- Late Arrivals

Checkout Queue

Booking	Customer	Product	Time	Qty	Readiness	Status
RNT-10482	John Ahmed	Mountain Bike	14:00	2	100%	Ready
RNT-10483	Sara Ali	Double Kayak	14:15	1	80%	Action Required
RNT-10484	ABC Group	Bikes	14:30	12	92%	Preparing

Filters

- Location
- Product
- Start Time
- Booking Type
- Group/Individual
- Readiness
- Status

Actions

Scan Voucher | Search Booking | Walk-In Checkout

Walk-in checkout should use the same underlying reservation process rather than create a completely separate operational transaction.

Screen 2 — Voucher Scan & Reservation Retrieval

Purpose

Quickly identify the customer reservation when they arrive.

The original requirement specifically requires scanning and validating the customer's voucher/ticket.

Scan Methods

- QR Voucher
- Barcode
- Mobile Ticket
- Wallet Credential
- Manual Booking Number
- Customer Search

Scan Result

✓ VALID RESERVATION

Booking:

RNT-20260908-10482

Customer:

John Ahmed

Product:

Mountain Bike – Adult × 2

Scheduled:

14:00–16:00

Location:

North Station

Validation

Automatically check:

- Correct venue
- Correct rental location
- Correct date
- Valid time window
- Reservation status
- Quantity
- Payment
- Deposit
- Agreement
- Participant requirements

If invalid:

✗ Reservation cancelled.

or:

⚠ Reservation is scheduled for 16:00. Early checkout requires authorization.

Screen 3 — Checkout Readiness Validation

Purpose

Prevent equipment from being handed over until mandatory requirements are satisfied.

Readiness Checklist

Reservation

- ✓ Booking confirmed
- ✓ Correct location
- ✓ Rental period valid

Customer

- ✓ Customer details complete
- ✓ ID verified
- ✓ Age requirement satisfied

Participants

- ✓ Participant details complete
- ✓ Guardian consent where required

Commercial

- ✓ Rental payment complete
- ✓ Deposit authorized

Agreement

- ✓ Rental agreement signed
- ✓ Liability waiver signed
- ✓ Safety declaration accepted

Inventory

- ✓ Required quantity available

Result

READY FOR EQUIPMENT ASSIGNMENT

or:

ACTION REQUIRED

Example:

- ⚠ Deposit authorization failed.

Actions:

Retry Deposit | Change Payment Method | Supervisor Override

Any override must be permission-controlled and audited.

Screen 4 — Equipment Assignment Workspace

Purpose

Assign the actual physical equipment to the reservation.

This is the point where:

2 Mountain Bikes

becomes:

BIKE-017 + BIKE-028

Reservation

Mountain Bike – Adult

Required:

2

Assigned:

0 / 2

Available Equipment

Asset	Condition	Location	Usage	Status
BIKE-017	Excellent	North	82 hrs	Available
BIKE-028	Good	North	64 hrs	Available
BIKE-031	Fair	North	121 hrs	Available

Assignment Methods

- Scan QR
- Scan Barcode
- RFID/NFC where supported
- Manual Selection
- Auto-Suggest

AI Recommendation

Recommended assets: BIKE-017 and BIKE-028.

Both are in good condition and have lower accumulated usage than other available assets.

This can help balance asset utilization.

Screen 5 — Equipment Scan & Validation

Purpose

Ensure the operator cannot accidentally assign the wrong, unavailable or unsafe equipment.

Scan

[SCAN EQUIPMENT QR]

Result:

BIKE-017

Product:

Mountain Bike – Adult

Location:

North Station

Status:

AVAILABLE

Condition:

GOOD

Last Inspection:

Today 09:30

Maintenance:

No active maintenance

Validation Result

✓ ELIGIBLE FOR RENTAL

If operator scans:

BIKE-022

Result:

✗ CANNOT ASSIGN

Reason:

Equipment is under maintenance.

Other rejection reasons:

- Wrong product

- Wrong location
- Already rented
- Reserved for another purpose
- Inspection required
- Damaged
- Out of service
- Lost
- Retired

This uses the asset status established in Board 2.

Screen 6 — Pre-Rental Condition Inspection

Purpose

Document the equipment condition before handover.

Photo capture at checkout is specifically recommended in the source.

Inspection Checklist

For a bicycle, for example:

- Frame
- Tires
- Brakes
- Seat
- Chain
- Pedals
- Helmet
- Accessories

Each:

Good | Fair | Existing Damage | Unsafe

Existing Damage

Example:

Small scratch — right frame.

Photo Evidence

Support:

Take Photo | Upload Photo

Photos should be timestamped and linked to:

- Asset
- Reservation
- Operator
- Checkout inspection

Customer Acknowledgment

Optional configurable setting:

Customer confirms equipment condition at handover.

Customer Signature / Accept

This significantly reduces later damage disputes.

Screen 7 — Safety & Handover Checklist

Purpose

Ensure mandatory operational and safety steps are completed before starting the rental.

Example — Kayak

- ✓ Life jacket provided
- ✓ Paddle provided
- ✓ Safety briefing completed
- ✓ Emergency procedure explained
- ✓ Restricted areas explained
- ✓ Customer confirms swimming ability

Example — Bicycle

- ✓ Helmet provided
- ✓ Brake check completed
- ✓ Seat adjusted
- ✓ Safety briefing completed

The checklist should be configurable by rental product/category from Board 1.

Staff Confirmation

Safety briefing completed by:

Operator – Ahmed

Time:

13:56

Customer Confirmation

I confirm that I received the equipment and required safety instructions.

Accept / Signature

Screen 8 — Deposit & Financial Handover Validation

Purpose

Give the rental operator a clear commercial status without requiring them to enter the Finance module.

Booking Charges

Rental:

AED 190 — PAID

Security Deposit

Required:

AED 400

Method:

Credit Card Pre-Authorization

Status:

✓ **AUTHORIZED**

Authorization Reference:

PA-882103

Possible States

- Not Required
- Pending
- Authorized
- Collected
- Failed
- Waived
- Partially Authorized

If failed:

CHECKOUT BLOCKED

unless authorized override rules permit continuation.

Cash Deposit

If cash is used:

Cash Deposit Received: AED 400

Staff:

Ahmed

Drawer:

POS-04

Financial execution remains within TICVAI Payments/Finance integrations.

Screen 9 — Group & Multi-Item Checkout

Purpose

Provide an efficient checkout workflow for group rentals and reservations containing multiple equipment units.

This is particularly important because the original requirements support group reservations.

Group Booking

ABC Tourism

Participants:

12

Mountain Bikes:

12

Assignment Grid

Participant	Asset	Inspection	Waiver	Status
Ahmed	BIKE-017	✓	✓	Ready
John	BIKE-018	✓	✓	Ready
Sara	BIKE-021	✓	✓	Ready
Omar	—	—	✓	Pending

Bulk Operations

- Scan next asset
- Auto-assign participant
- Assign equipment without participant
- Bulk inspection
- Print assignment list

Progress

11 / 12 Equipment Assigned

12 / 12 Waivers Complete

Checkout

The system should support:

Checkout All

or, where policy permits:

Partial Group Checkout

Example:

11 customers start now while the final participant completes requirements.

Screen 10 — Checkout Confirmation & Rental Activation

Purpose

Officially start the rental and transition the reservation into an active rental.

Final Summary

Booking: RNT-20260908-10482

Customer:

John Ahmed

Product:

Mountain Bike × 2

Assigned:

BIKE-017

BIKE-028

Scheduled Start:

14:00

Actual Checkout:

13:58

Expected Return:

15:58 / 16:00 according to configured policy

Deposit:

✓ Authorized

Inspection:

✓ Complete

Agreement:

✓ Signed

Safety:

✓ Complete

Action

START RENTAL

Upon confirmation, TICVAI must automatically:

1. Change reservation status to **Checked Out / Active Rental**.
2. Change assigned assets to **Rented**.
3. Record actual checkout time.
4. Record operator.
5. Record serial numbers.
6. Store inspection evidence.
7. Start rental/late-return monitoring.
8. Calculate expected return.
9. Update real-time availability.
10. Create complete audit records.

These checkout records directly implement the requirement to capture checkout time, staff member, assigned serial number and deposit status.

Customer Confirmation

Send:

Rental Started

Expected return:

16:00

Options:

Email | SMS | App Notification

The customer can receive:

- Assigned equipment
- Expected return time
- Return location

- Extension instructions
 - Rental terms
 - Support/contact information
-

Important Recommendation — Assignment Timing

One architectural point should be made very clear to Softlabs.

For most products:

Reservation Stage — Board 5

Customer reserves:

2 Mountain Bikes

No serial number necessarily needs to be assigned.

Checkout Stage — Board 6

Operator scans:

BIKE-017 + BIKE-028

Those assets become physically linked to the customer.

This avoids locking individual assets days or weeks before the customer arrives.

However, TICVAI should support an exception for **resource-specific booking**.

For example:

Cabana #C12

or:

Locker #L304

may be assigned during reservation if the business requires customers to select a specific resource.

Important Recommendation — Pre-Rental Evidence

I strongly recommend that the condition inspection be treated as a proper record rather than only a free-text note.

For every serialized asset, TICVAI should retain:

Condition Before Rental

-

Checkout Photos

-

Customer Acknowledgment

and Board 8 will later capture:

Condition After Rental

-

Return Photos

This enables:

BEFORE vs AFTER

and gives the damage workflow objective evidence.

Important Recommendation — Walk-In Customers

Board 6 should support walk-ins, but we should **not create a second rental architecture for them.**

The flow should be:

Walk-In Customer

↓

Quick Reservation

↓

Availability Check — Board 3

↓

Price/Deposit — Board 4

↓

Customer/Agreement

↓

Immediate Checkout — Board 6

This means both advance reservations and walk-ins ultimately use the same backend services.

Board 6 — Core Operational Flow

Customer Arrives

↓

Scan QR / Find Reservation

↓

Validate Booking

↓

Check Customer / Waiver / Payment / Deposit

↓

Select Equipment

↓

Scan Serial Number

↓

Validate Equipment

↓

Pre-Rental Inspection + Photos

↓

Safety Handover

↓

Customer Acknowledgment

↓

START RENTAL



Asset = RENTED



Reservation = ACTIVE RENTAL



BOARD 7 — ACTIVE RENTAL OPERATIONS

Integration Boundaries

Board 6 owns

- Customer arrival processing
- Reservation validation for checkout
- Physical equipment assignment
- Asset scan
- Pre-rental inspection
- Checkout evidence
- Safety handover
- Checkout confirmation
- Rental activation

Board 2 owns

Equipment master, serial numbers, location, condition and current operational status.

Board 3 owns

Availability calculations.

Board 4 owns

Deposit and commercial policy.

Board 5 owns

Reservation/customer/participant/waiver records.

Payments owns

Deposit authorization and financial transactions.

Board 7 owns

Everything occurring **after the rental starts and before return**.

Board 8 owns

Return, post-rental inspection, damage and final settlement.

Board 6 — Acceptance Criteria

Board 6 is complete when TICVAI can:

1. Display a real-time rental checkout queue.
2. Retrieve reservations through QR/barcode scanning or manual search.
3. Validate reservation date, time, location and operational status.

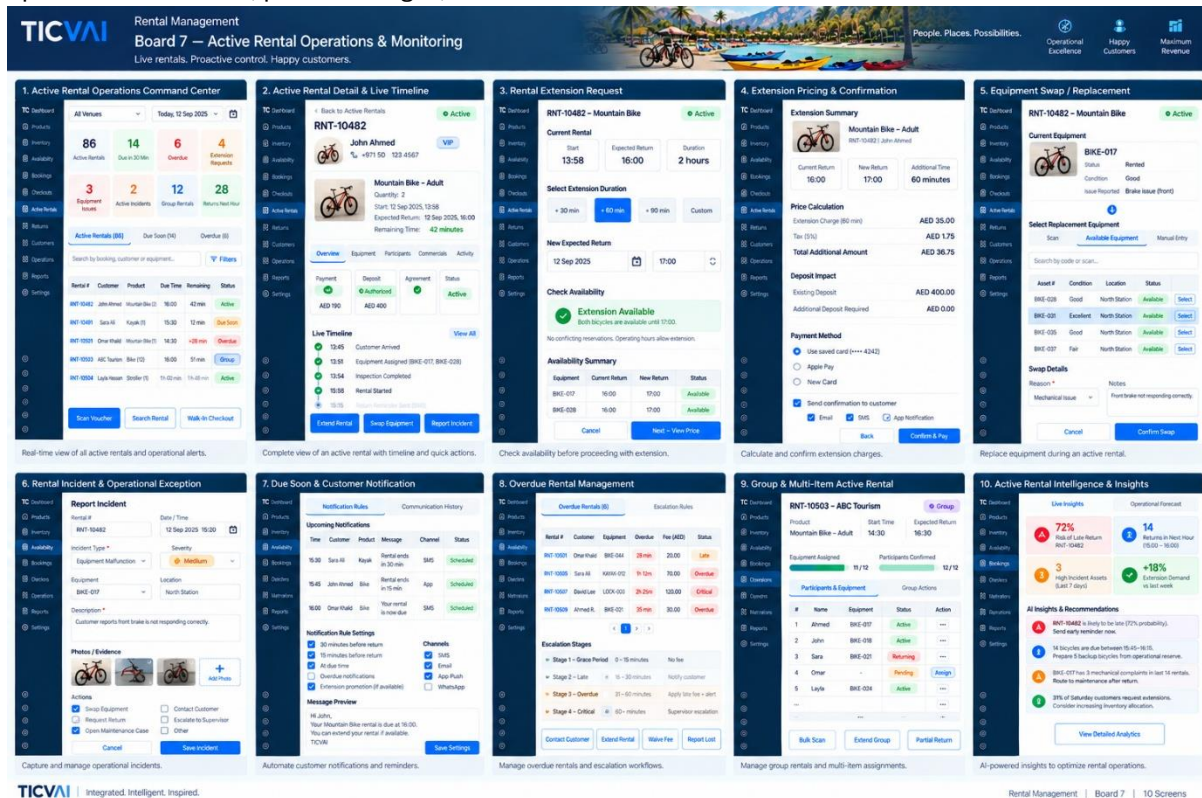
4. Validate customer, participant, agreement, payment and deposit readiness.
5. Assign actual serialized or pooled rental equipment.
6. Scan and validate equipment using QR/barcode and support RFID/NFC where applicable.
7. Prevent assignment of rented, damaged, maintenance, lost, unsafe or otherwise ineligible equipment.
8. Perform configurable pre-rental condition inspections.
9. Capture checkout photos and existing-damage evidence.
10. Capture product-specific safety/handover checklists.
11. Support customer acknowledgment/signature at handover.
12. Validate deposit/security-hold status before equipment release.
13. Support efficient group and multi-item assignment.
14. Support controlled partial group checkout.
15. Record **actual checkout time, operator, assigned serial numbers and deposit status.**
16. Automatically change assigned equipment to **Rented.**
17. Immediately update Rental Availability after checkout.
18. Calculate and communicate expected return time.
19. Support walk-in rentals through the same centralized rental architecture.
20. Maintain a complete auditable checkout record and pass the transaction to **Board 7 — Active Rental Operations.**

This makes Board 6 the critical **physical handover control point**: nothing leaves the rental location until TICVAI knows **who has it, which exact asset they have, its condition when it left, the financial/security status, and when it is expected back.**

Board 7 — Active Rental Operations & Monitoring

Module: 7 — Rental Management

Board Objective: Manage everything that happens **after equipment has been checked out and before it is returned**, including live rental monitoring, extensions, equipment swaps, overdue rentals, customer notifications, operational incidents, partial changes, and staff interventions.



Board 6 ended when:

Customer + Equipment + Checkout = Active Rental

Board 7 answers:

“What is happening while the rental is active, and what do we do when the customer wants more time, equipment needs to be replaced, or the rental becomes overdue?”

This board includes several of the enhancements I recommended earlier—particularly **extensions, equipment swaps/replacements, proactive overdue monitoring, customer communication, and operational exception handling.**

Screen 1 — Active Rental Operations Command Center

Purpose

Provide operators and supervisors with a real-time view of every rental currently in progress.

KPI Cards

- Active Rentals
- Due Within 30 Minutes
- Overdue
- Extensions Requested
- Equipment Swap Required
- Active Incidents
- Group Rentals Active

- Expected Returns Next Hour

Live Rental Table

Rental	Customer	Product	Equipment	Checkout	Due	Remaining	Status
RNT-10482	John Ahmed	Bike x2	017, 028	14:00	16:00	42 min	Active
RNT-10491	Sara Ali	Kayak	K-022	13:00	15:30	12 min	Due Soon
RNT-10501	Omar Khalid	Bike	B-044	12:30	14:30	+28 min	Overdue

Status Indicators

Active

Due Soon

Extension Pending

Extended

Overdue

Incident

Return in Progress

Filters

- Venue
- Location
- Product
- Customer
- Rental status
- Expected return
- Overdue duration
- Group/Individual

The screen should automatically refresh so operators can manage the live rental operation without manually searching transactions.

Screen 2 — Active Rental Detail & Live Timeline

Purpose

Provide the complete operational view of one active rental.

Rental Header

RNT-20260908-10482

Customer:

John Ahmed

Status:

ACTIVE

Rental Information

Product:

Mountain Bike ×2

Equipment:

BIKE-017

BIKE-028

Actual Checkout:

13:58

Expected Return:

16:00

Remaining:

42 Minutes

Commercial Status

Rental Paid:

✓ AED 190

Deposit:

✓ AED 400 Authorized

Tabs

Overview | Equipment | Participants | Commercials | Activity | Communications | Audit

Live Timeline

13:45 — Customer Arrived

13:51 — Equipment Assigned

13:54 — Inspection Completed

13:58 — Rental Started

15:15 — Return Reminder Scheduled

Quick Actions

Extend Rental

Swap Equipment

Report Incident

Contact Customer

Start Return

Screen 3 — Rental Extension Request

Purpose

Allow the customer or operator to request additional rental time.

This is one of the important capabilities I recommend formally adding to the rental scope.

Current Rental

Start:

14:00

Expected Return:

16:00

Current Duration:

120 Minutes

Extension Request

Additional Duration:

+30 min

+60 min

+90 min

or:

Custom

Requested New Return

17:00

Before accepting the extension, TICVAI must call:

Board 3 — Availability Engine

The system must check whether extending this rental will conflict with:

- Future reservations
- Maintenance
- Operating hours
- Inventory requirements
- Turnaround time
- Protected inventory

Result

✓ **EXTENSION AVAILABLE**

or:

✗ **EXTENSION NOT AVAILABLE**

Reason:

BIKE-017 is required for another reservation at 16:30.

Screen 4 — Extension Pricing & Confirmation

Purpose

Calculate the commercial impact of an approved extension.

Once Board 3 confirms availability, Board 4 calculates the extension charge.

Example

Current Return:

16:00

New Return:

17:00

Additional Time:

60 Minutes

Price

Standard Extension:

AED 35

Tax:

AED 1.75

Additional Amount

AED 36.75

Deposit

Existing Deposit:

AED 400

Additional Deposit Required:

AED 0

Customer Confirmation

Your rental will be extended until 17:00 for AED 36.75.

Actions:

Accept & Pay

Decline

After Confirmation

TICVAI automatically:

- Updates expected return
 - Updates reservation duration
 - Charges additional amount
 - Updates Board 3 availability
 - Updates future capacity
 - Records extension history
 - Sends confirmation to customer
-

Screen 5 — Equipment Swap / Replacement

Purpose

Replace equipment during an active rental without terminating and recreating the rental.

This is another capability I strongly recommend adding.

Example:

Customer reports:

BIKE-017 has a brake problem.

Current Equipment

BIKE-017

Status:

Rented

Condition:

Good at Checkout

Replacement

Scan:

BIKE-035

System validates:

- ✓ Same/compatible product
- ✓ Correct location
- ✓ Available
- ✓ Safe
- ✓ No maintenance block

Swap Reason

- Mechanical Issue
- Damage
- Customer Comfort
- Incorrect Size
- Safety Concern
- Staff Decision
- Other

Workflow

BIKE-017

Rented

↓

SWAP

↓

BIKE-035

Rented

The original asset should not automatically become available.

Instead:

BIKE-017 → Inspection Required

This prevents potentially faulty equipment from being immediately rented to another customer.

Screen 6 — Rental Incident & Operational Exception

Purpose

Capture incidents occurring while equipment is in customer possession.

Incident Types

- Equipment Malfunction
- Customer Accident
- Equipment Damage
- Lost Equipment
- Lost Accessory
- Safety Incident
- Customer Complaint
- Location Issue
- Weather/Safety Closure
- Other

Incident Record

Rental:

RNT-10482

Equipment:

BIKE-017

Incident:

Equipment Malfunction

Severity:

Medium

Description:

Customer reports front brake is not responding correctly.

Evidence

- Photos
- Video
- Documents
- Staff Notes

Actions

Swap Equipment

Request Return

Escalate

Open Maintenance Case

Contact Supervisor

For serious incidents, TICVAI should support escalation according to configured roles and workflows.

Screen 7 — Due Soon & Customer Notification Management

Purpose

Proactively communicate with customers before their rental becomes overdue.

The original matrix requires overdue rental notifications.

Notification Rules

Example:

30 minutes before return

Your Mountain Bike rental is due at 16:00.

15 minutes before return

Your rental ends in 15 minutes. Return to North Station or extend your rental if available.

At due time

Your rental is now due for return.

Channels

- SMS
- Email
- Mobile App Push
- WhatsApp where integrated

Customer Actions

Where enabled, the notification can provide:

Return Directions

Request Extension

Contact Rental Desk

Communication History

Time	Message	Channel	Status
15:30	Return Reminder	SMS	Delivered
15:45	Return Reminder	App	Read

Screen 8 — Overdue Rental Management

Purpose

Provide operators with a dedicated workflow for rentals that have exceeded their expected return time.

The original requirement calls for automatic late-fee calculation, configurable grace periods and overdue notifications.

Overdue Dashboard

Rental	Customer	Equipment	Due	Overdue	Fee	Contact
RNT-10501	Omar	B-044	14:30	28 min	AED 20	Contacted
RNT-10478	Sara	K-012	13:45	1h 12m	AED 70	No Response

Overdue Stages

I recommend configurable escalation levels.

Stage 1 — Grace Period

0–15 minutes
No fee.

Stage 2 — Late

16–30 minutes
Customer notification.

Stage 3 — Overdue

31–60 minutes
Late fee + operator alert.

Stage 4 — Critical

60+ minutes
Supervisor escalation.

Actions

Contact Customer

Extend Rental

Waive Fee Request

Report Lost

Escalate

Late-fee amounts must come from Board 4 rather than being independently configured here.

Screen 9 — Active Group Rental Management

Purpose

Manage group rentals where multiple pieces of equipment may have different operational states.

Example:

ABC Tourism

12 Participants
12 Mountain Bikes

Equipment Status

Participant	Equipment	Due	Status
Ahmed	BIKE-017	16:00	Active
John	BIKE-018	16:00	Active
Sara	BIKE-021	16:00	Returning
Omar	BIKE-023	16:00	Swap Required
Layla	BIKE-024	16:00	Active

Group Actions

- Extend Entire Group
- Extend Selected Participants
- Swap Equipment
- Contact Group Leader
- Start Partial Return
- Report Incident

Recommended Capability — Partial Return

The system should allow:

12 rented

↓

5 returned

↓

7 still active

The group rental remains:

PARTIALLY RETURNED

until all equipment is returned or otherwise resolved.

This is important for group rentals and multi-item transactions.

Screen 10 — Active Rental Intelligence & Operational Alerts

Purpose

Use AI and real-time operational data to identify risks before they become operational problems.

Live Operational Intelligence

Display:

- Rentals due next hour
- Predicted late returns
- Future reservation conflicts
- Extension demand
- Equipment incident trends
- Location return pressure
- Expected inventory shortage

AI Late Return Prediction

Example:

RNT-10482

Expected Return: 16:00

Historical/customer/operational indicators suggest a **72% probability of late return**.

BIKE-017 is required for another reservation at 16:30.

Recommended Action

Send early return reminder now.

AI Capacity Risk

14 bicycles are due between 15:45–16:15.

11 new bicycle rentals are scheduled between 16:15–16:45.

Risk: delayed returns may affect upcoming reservations.

Recommended:

Prepare 5 backup bicycles from operational reserve.

AI Incident Pattern

BIKE-017 has received 3 mechanical complaints in the last 14 rentals.

Recommendation:

Route BIKE-017 to maintenance after return.

AI Extension Insight

31% of Saturday kayak customers request extensions.

Consider adjusting standard rental durations or increasing inventory allocation during 15:00–19:00.

AI should recommend actions, but operational decisions remain controlled by authorized staff.

Important Rental Status Model

I recommend that Softlabs implement a clear rental lifecycle rather than relying on a generic booking status.

Reservation

Confirmed

↓

Checkout

Ready for Checkout

↓

Rental Started

Active

↓

Possible states:

Active

Extended

Swap in Progress

Incident

Due Soon

Overdue

Partially Returned

↓

Return

Return in Progress

↓

Board 8

Completed / Completed with Charges / Lost / Damage Case

This separation will make reporting and operations significantly cleaner.

Important Extension Architecture

Extensions must never simply change the return timestamp.

The correct workflow is:

Customer Requests Extension

↓

Check Board 3 Availability

↓

If available:

Board 4 Calculates Extension Price

↓

Customer Accepts

↓

Payment

↓

Update Rental

↓

Update Availability

↓

Notify Customer

If unavailable:

Extension declined because future capacity is already committed.

This prevents an extension from causing an overbooking later in the day.

Important Equipment Swap Architecture

A swap should preserve one rental transaction.

For example:

Rental RNT-10482

Original:

BIKE-017

14:00–15:10

Reason:

Mechanical issue

Replacement:

BIKE-035

15:10–16:00

The complete rental history should therefore show:

BIKE-017 → BIKE-035

rather than creating a new reservation.

This also ensures both assets have correct usage and maintenance history.

Board 7 — Core Operational Flow

Rental Active

↓

Monitor Time & Equipment

↓

Possible:

Normal Operation

or

Extension Request

or

Equipment Swap

or

Incident

or

Overdue

↓

Customer Notification / Operator Action

↓

Customer Returns Equipment

↓

BOARD 8 — RETURN, DAMAGE & SETTLEMENT

Integration Boundaries

Board 7 owns

- Active rental monitoring
- Rental extension workflow
- Equipment swaps
- Active rental incidents
- Due-soon monitoring
- Overdue management
- Customer operational notifications
- Active group rental management
- Partial-return initiation
- Operational AI alerts

Board 2 owns

Physical equipment master and asset lifecycle.

Board 3 owns

Availability validation for extensions and future capacity.

Board 4 owns

Extension and late-fee calculations.

Board 5 owns

Original reservation.

Board 6 owns

Original equipment checkout and handover.

Board 8 owns

Physical return, final inspection, damage assessment and settlement.

Board 9 owns

Detailed maintenance workflow created from equipment incidents.

Board 7 — Acceptance Criteria

Board 7 is complete when TICVAI can:

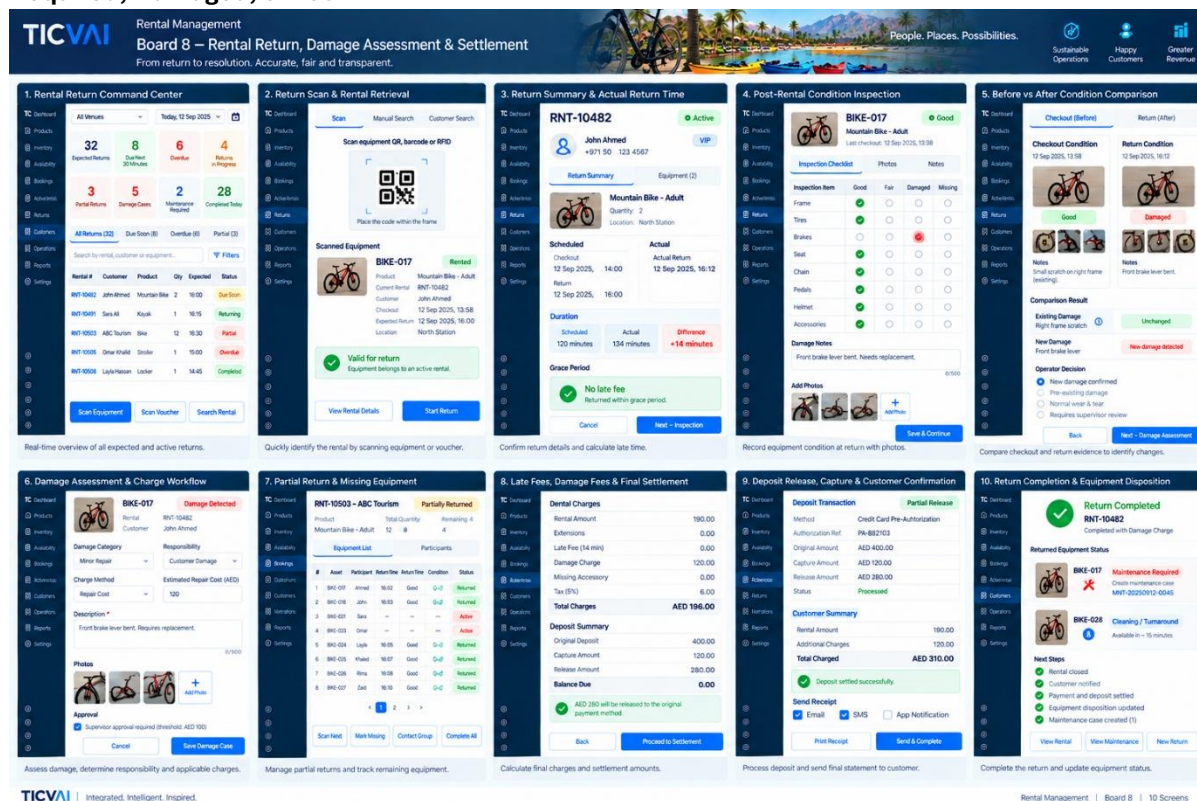
1. Monitor all active rentals in real time.
2. Identify rentals due soon, overdue or requiring operational intervention.
3. Display complete active-rental details and timeline.
4. Support rental extension requests.
5. Validate extension availability through Board 3 before approval.
6. Calculate extension pricing through Board 4.
7. Collect additional payment where required.
8. Automatically update availability after an extension.
9. Replace/swap equipment without creating a new rental.
10. Route replaced/problematic equipment to **Inspection Required** or Maintenance rather than automatically making it available.
11. Capture active-rental incidents with notes, photos and evidence.
12. Trigger maintenance cases from equipment incidents.
13. Automatically send configurable due-soon and overdue notifications.
14. Apply grace periods and retrieve automatic late fees from Board 4.
15. Support configurable overdue escalation.
16. Support group-level and participant/equipment-level active rental operations.
17. Support partial returns for multi-item/group rentals.
18. Maintain complete equipment usage history across swaps.
19. Provide AI-driven predictions for late returns, future reservation conflicts, inventory pressure and equipment issues.
20. Pass returned equipment and associated incidents cleanly into **Board 8 — Return, Damage & Settlement**.

This makes Board 7 the **live operational control tower for every rental currently outside the rental counter**—giving TICVAI visibility and control from the moment equipment leaves until it comes back.

Board 8 — Rental Return, Damage Assessment & Settlement

Module: 7 — Rental Management

Board Objective: Manage the complete physical and financial return process: identify the active rental, scan returned equipment, record actual return time, perform post-rental inspection, compare condition against checkout evidence, manage partial returns, assess damage/loss, calculate applicable charges, settle/release deposits, and determine whether equipment becomes **Available**, **Cleaning/Inspection Required**, **Maintenance Required**, **Damaged**, or **Lost**.



The original requirements require TICVAI to scan the rented item, retrieve the reservation, record the return time, condition, damage notes and optional photos, and classify the outcome as **Returned**, **Returned Damaged**, **Lost**, or **Maintenance Required**.

The source also requires automatic late-fee calculation and configurable grace periods, while deposits must support full refund/release, partial refund and forfeiture.

Board 8 answers:

“The equipment has come back. What came back, when did it return, what condition is it in, what does the customer owe, what happens to the deposit, and can the equipment be rented again?”

I recommend exactly **10 screens**.

Screen 1 — Rental Return Command Center

Purpose

Provide operators with a live operational view of all expected, in-progress, partial, overdue and completed returns.

KPI Cards

- Expected Returns Today
- Due Next 30 Minutes
- Return in Progress
- Overdue
- Partial Returns

- Damage Cases
- Maintenance Required
- Returns Completed Today
- Deposits Pending Settlement
- Equipment Awaiting Inspection

Return Queue

Rental	Customer	Product	Qty	Expected	Status
RNT-10482	John Ahmed	Mountain Bike	2	16:00	Due Soon
RNT-10491	Sara Ali	Kayak	1	16:15	Returning
RNT-10503	ABC Tourism	Bikes	12	16:30	Partial Return
RNT-10505	Omar Khalid	Stroller	1	15:00	Overdue

Filters

- Venue
- Return Location
- Product
- Expected Return
- Rental Status
- Damage Status
- Deposit Status
- Group/Individual

Actions

Scan Equipment

Scan Rental Voucher

Search Rental

Start Return

Screen 2 — Return Scan & Rental Retrieval

Purpose

Quickly identify which active rental the returned equipment belongs to.

The original requirements specifically require scanning the rented item and retrieving its reservation.

Scan

[SCAN QR / BARCODE / RFID]

Example result:

BIKE-017

Product:

Mountain Bike – Adult

Status:

RENTED

Current Rental:

RNT-20260908-10482

Customer:

John Ahmed

Checkout:

13:58

Expected Return:

16:00

Return Location

North Station

Validation

System checks:

- Equipment is currently rented
- Equipment belongs to this rental
- Return location is permitted
- Rental is active
- Asset identity is valid

Result

✓ **VALID FOR RETURN**

Actions:

Start Return

or:

Add to Existing Return

Screen 3 — Return Summary & Actual Return Time

Purpose

Establish the operational return event before inspection and financial settlement.

Rental

RNT-10482

Customer:

John Ahmed

Equipment

BIKE-017

BIKE-028

Scheduled

Checkout:

14:00

Expected Return:

16:00

Actual

Return Scan:

16:12

Duration

Scheduled:

120 minutes

Actual:

134 minutes

Difference:

+14 Minutes

Grace Period

15 Minutes

Result:

✓ **NO LATE FEE**

If returned at 16:28:

Overdue:

28 minutes

Grace:

15 minutes

Chargeable Late Time:

13 minutes

Board 4 then provides the applicable late-fee calculation.

The original requirement requires capture of the actual return time.

Screen 4 — Post-Rental Condition Inspection

Purpose

Perform a structured inspection of returned equipment.

Asset

BIKE-017

Condition Before:

Good

Return Inspection

Inspection Item	Good	Fair	Damaged	Missing
Frame	●	○	○	—
Tires	●	○	○	—
Brakes	○	○	●	—

Inspection Item	Good	Fair	Damaged	Missing
Seat	●	○	○	—
Chain	●	○	○	—
Helmet	●	○	○	○
Accessories	●	○	○	○

Damage Notes

Front brake lever bent.

Photo Evidence

Take Photos

Support multiple photos.

The source explicitly supports damage notes and optional photos during return.

Inspection Outcome

- Passed
- Cleaning Required
- Minor Damage
- Major Damage
- Maintenance Required
- Unsafe
- Missing/Lost

Screen 5 — Before vs After Condition Comparison

Purpose

Use the evidence captured during Board 6 checkout to establish whether damage existed before the rental or occurred during it.

This is one of the important capabilities I recommended adding.

Asset

BIKE-017

BEFORE CHECKOUT

Condition:

Good

Photos:

[Photo 1] [Photo 2] [Photo 3]

Existing Damage:

Small scratch on right frame.

AFTER RETURN

Condition:

Damaged

Photos:

[Photo 1] [Photo 2] [Photo 3]

New Finding:

Front brake lever bent.

Comparison

Existing Damage

✓ Right frame scratch

New Damage

⚠ Front brake lever

Operator Decision

New Damage Confirmed

or:

Pre-Existing

or:

Normal Wear & Tear

or:

Requires Supervisor Review

This evidence is particularly important when a customer charge is proposed.

Screen 6 — Damage Assessment & Charge Workflow

Purpose

Convert confirmed equipment damage into a controlled operational and commercial case.

The source specifically recommends a formal **damage assessment workflow**.

Damage Case

Case:

DMG-20260908-0021

Rental:

RNT-10482

Asset:

BIKE-017

Damage Category

- Cosmetic
- Minor Repair
- Major Repair
- Safety Critical
- Missing Component

- Total Loss

Responsibility

- Customer Damage
- Normal Wear
- Pre-Existing
- Operational Damage
- Undetermined

Charge Method

Support:

Fixed Damage Fee

Repair Cost

Replacement Cost

Manual Assessment

No Charge

Example

Damage:

Brake lever

Estimated Repair:

AED 120

Customer Charge:

AED 120

Approval

If charge exceeds configured threshold:

Supervisor approval required.

Evidence

- Before photos
- After photos
- Inspection notes
- Customer acknowledgment
- Operator notes

Screen 7 — Partial Return & Missing Equipment

Purpose

Handle multi-item and group rentals where equipment is returned at different times.

This implements the partial-return capability we added in Board 7.

Group Rental

ABC Tourism

Original:

12 Bikes

Returned:

8 / 12

Still Active:

4

Equipment

Asset	Participant	Return	Condition	Status
BIKE-017	Ahmed	16:02	Good	Returned
BIKE-018	John	16:03	Good	Returned
BIKE-021	Sara	—	—	Active
BIKE-023	Omar	—	—	Active

Rental Status

PARTIALLY RETURNED

Actions:

Return Selected Assets

Contact Group Leader

View Remaining Equipment

Missing Equipment

If an expected item is not returned:

Mark Missing

or:

Keep Rental Active

or:

Escalate as Lost

The overall rental remains open until all equipment is resolved.

Screen 8 — Late Fees, Damage Fees & Final Settlement

Purpose

Calculate the customer's final rental financial position.

Board 8 should **consume**, not recreate, the commercial policies configured in Board 4.

Rental Summary

Rental Charge:

AED 190

Paid:

AED 190

Additional Charges

Late Fee:

AED 0

Damage:

AED 120

Missing Accessory:

AED 25

Extension:

AED 0

Total Additional Charges

AED 145

Deposit

Security Hold:

AED 400

Settlement

Capture from Deposit:

AED 145

Release:

AED 255

Customer Additional Payment

AED 0

If charges exceed deposit:

Deposit:

AED 400

Final Charges:

AED 550

Capture Deposit:

AED 400

Additional Amount Due

AED 150

Payment processing must use TICVAI Payments.

Screen 9 — Deposit Release, Capture & Customer Confirmation

Purpose

Complete the deposit lifecycle and provide the customer with a transparent final statement.

The source requires full refund/release, partial refund and deposit forfeiture, together with transaction history.

Deposit

Original:

AED 400

Method:

Credit Card Pre-Authorization

Settlement Options

Successful Return

Release Full Deposit

AED 400

Partial Capture

Damage:

AED 120

Capture AED 120

Release AED 280

Full Capture / Forfeiture

Lost equipment:

Capture AED 400

Approval

Where required:

Supervisor Approval

Customer Final Statement

Rental:

AED 190

Additional Charges:

AED 120

Deposit Captured:

AED 120

Deposit Released:

AED 280

Delivery

Email | SMS | App | Print

Customer should receive a clear explanation of every final charge.

Screen 10 — Return Completion & Equipment Disposition

Purpose

Complete the rental while determining exactly what happens to every returned physical asset.

This is extremely important because **returning an item does not necessarily mean it becomes immediately available again.**

Equipment Disposition

BIKE-028

Inspection:

Passed

Turnaround:

15 Minutes

Next Status:

CLEANING / TURNAROUND

then:

AVAILABLE

BIKE-017

Inspection:

Damage Detected

Next Status:

MAINTENANCE REQUIRED

Create Maintenance Case:

MNT-20260908-0098

Rental Completion

All equipment resolved:

✓

Financial settlement:

✓

Deposit processed:

✓

Inspection complete:

✓

Damage case:

✓

Customer notified:

✓

Final Rental Status

COMPLETED WITH DAMAGE CHARGE

Other possible final statuses:

- Completed
- Completed with Late Fee
- Completed with Damage
- Completed with Charges
- Partially Returned
- Lost Equipment Case
- Settlement Pending

Actions

Complete Rental

View Final Statement

Open Maintenance Case

View Customer History

Important Recommendation — Do Not Make Returned Equipment Immediately Available

This should be explicitly written into the Softlabs scope.

A common mistake would be:

Rented → Returned → Available

I recommend:

Normal Return

Rented

↓

Returned

↓

Inspection

↓

Cleaning / Turnaround

↓

AVAILABLE

For damaged equipment:

Rented

↓

Returned

↓

Inspection

↓

MAINTENANCE REQUIRED

For unsafe equipment:

Rented

↓

Returned

↓

OUT OF SERVICE

This connects directly to the availability engine in Board 3.

Important Recommendation — Damage Evidence Chain

For every damage dispute, TICVAI should be able to reconstruct:

Before Rental

- Checkout inspection
- Checkout condition
- Photos
- Existing damage
- Customer acknowledgment
- Operator
- Timestamp

↓

During Rental

- Equipment swap
- Incident
- Customer communication

↓

After Rental

- Return inspection
- Return condition
- Photos
- New damage
- Operator
- Timestamp

↓

Assessment

- Damage classification
- Responsibility
- Charge
- Approval

↓

Settlement

- Deposit capture
- Refund/release
- Additional payment

This gives TICVAI a strong auditable evidence chain.

Important Recommendation — Damage Does Not Automatically Mean Customer Charge

We should make this clear.

A damaged item could be:

Customer-caused damage

→ Charge may apply.

Normal wear and tear

→ No customer charge.

Pre-existing damage

→ No customer charge.

Equipment failure

→ No customer charge and potentially maintenance/compensation workflow.

Undetermined

→ Supervisor review.

Therefore:

Damage Detection ≠ Automatic Customer Charge

The damage assessment workflow must determine responsibility before settlement.

Important Recommendation — Lost Equipment

Lost equipment should have its own controlled workflow.

Example

Customer does not return:

BIKE-017

Operator selects:

MARK AS LOST

TICVAI should:

1. Change asset status to **Lost**.
2. Remove it from availability.
3. Calculate applicable loss/replacement charge.
4. Evaluate deposit capture.
5. Create operational incident.
6. Require approval where configured.
7. Retain the rental/customer relationship.
8. Allow later recovery.

If BIKE-017 is found three days later:

Lost → Found → Inspection Required

It should not automatically return to Available.

Board 8 — Core Return Flow

Customer Returns

↓

Scan Equipment

↓

Retrieve Active Rental

↓

Record Actual Return

↓

Check Late Status



Post-Rental Inspection



Compare BEFORE vs AFTER



Possible:

Passed

or

Damage

or

Missing

or

Maintenance Required



Calculate Charges



Settle Deposit



Customer Confirmation



Equipment Disposition



COMPLETE RENTAL

Integration Boundaries

Board 8 owns

- Physical return
- Actual return timestamp
- Return inspection
- Before/after comparison
- Damage assessment
- Partial return
- Missing/lost return handling
- Final rental settlement orchestration
- Deposit release/capture request
- Equipment disposition
- Rental completion

Board 2 owns

Equipment master, serial numbers and long-term asset status/history.

Board 3 owns

Recalculated availability once equipment becomes eligible for rental again.

Board 4 owns

Late-fee, damage-fee/deposit and commercial policies.

Board 6 owns

Checkout condition and before-rental evidence.

Board 7 owns

Active rental incidents, extensions, swaps and overdue monitoring before return.

Payments owns

Actual authorization capture, additional charge and refund/release processing.

Finance owns

Financial posting, reconciliation and accounting.

Board 9 owns

Maintenance and repair cases generated from returned equipment.

Board 8 — Acceptance Criteria

Board 8 is complete when TICVAI can:

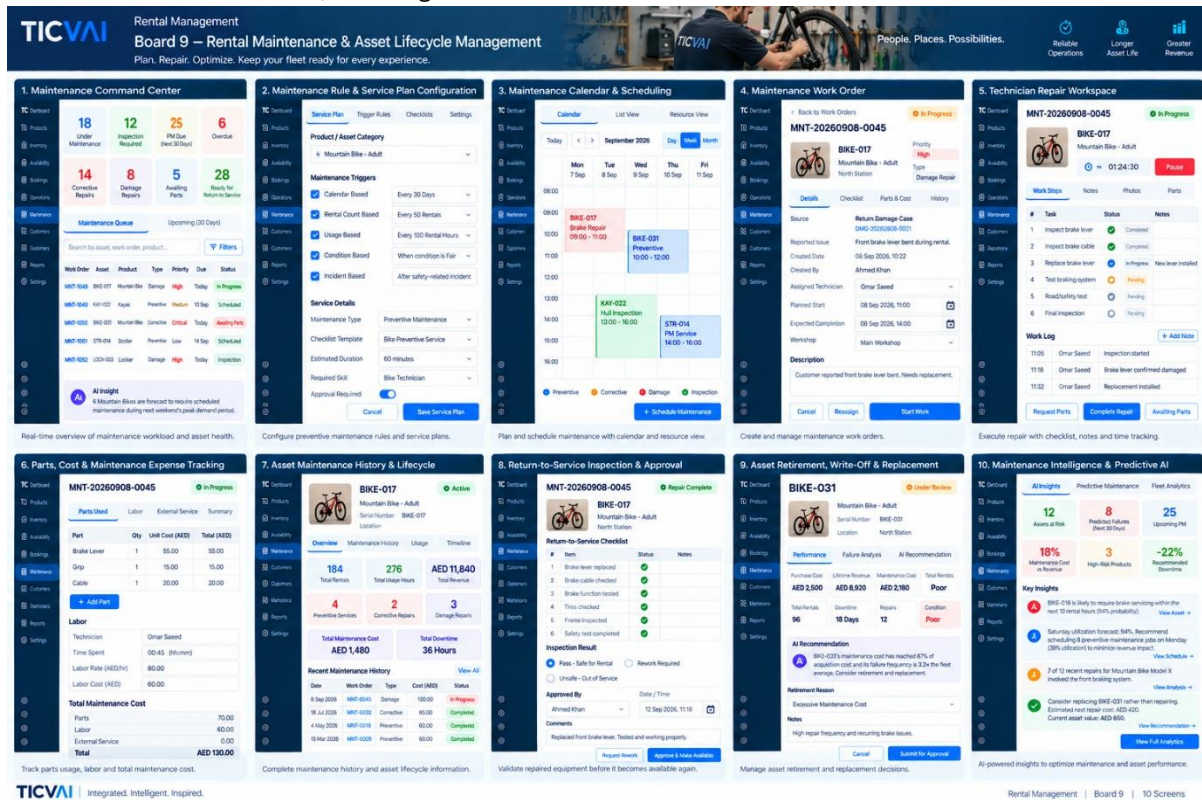
1. Provide a real-time return operations queue.
2. Retrieve active rentals by scanning rental equipment or customer voucher.
3. Validate equipment against the active rental.
4. Record exact actual return time.
5. Calculate overdue duration and apply configured grace periods.
6. Perform structured post-rental equipment inspection.
7. Capture return condition, notes and photo evidence as required by the original scope.
8. Compare pre-rental and post-rental condition/evidence.
9. Distinguish new damage from pre-existing damage and normal wear.
10. Create structured damage-assessment cases.
11. Support configurable damage categories and responsibility determination.
12. Support supervisor approval for applicable damage charges.
13. Support partial returns for group and multi-item rentals.
14. Track equipment that remains active after a partial return.
15. Handle missing and lost equipment without incorrectly closing the rental.
16. Calculate final late, damage, loss and other applicable charges through Board 4 policies.
17. Support full deposit release, partial capture/release and full capture/forfeiture.
18. Process financial transactions through TICVAI Payments rather than duplicating payment functionality.
19. Generate and deliver a transparent final customer statement.
20. Route each returned asset to the correct next state: **Turnaround, Available, Inspection Required, Maintenance Required, Out of Service, Damaged or Lost.**
21. Prevent returned equipment from becoming sellable until required inspection/turnaround is complete.
22. Automatically update Board 3 availability when the asset genuinely becomes available.
23. Maintain the complete **checkout → active rental → return → inspection → damage → settlement** audit trail.
24. Generate maintenance cases for applicable assets and pass them to **Board 9 — Maintenance & Asset Lifecycle Management.**

This makes Board 8 the point where TICVAI closes the operational and financial loop: **we know exactly what came back, when it came back, its condition compared with when it left, what the customer owes, what happens to the deposit, and whether that equipment is actually safe and ready to rent again.**

Board 9 — Rental Maintenance & Asset Lifecycle Management

Module: 7 — Rental Management

Board Objective: Manage the complete maintenance lifecycle of rental equipment—from inspection findings and damage cases through preventive maintenance, corrective repair, work orders, technician actions, downtime, return-to-service validation, and long-term asset health.



The original requirements require TICVAI to support marking rental items **Under Maintenance**, scheduling maintenance periods, maintaining maintenance history, preventing rental during maintenance, and supporting **Preventive Maintenance, Corrective Maintenance, and Damage Repair**.

The source also specifically recommends a maintenance scheduling module and AI-based utilization/demand forecasting.

Board 9 answers:

“Which rental assets need maintenance, why do they need it, who will repair them, how long will they remain unavailable, what did the repair cost, and when are they safe to rent again?”

I recommend exactly **10 screens**.

Screen 1 — Maintenance Command Center

Purpose

Provide maintenance teams and rental management with a real-time view of rental asset health and maintenance workload.

KPI Cards

- Assets Under Maintenance
- Inspection Required
- Preventive Maintenance Due
- Overdue Maintenance
- Corrective Repairs

- Damage Repairs
- Awaiting Parts
- Ready for Return to Service
- Average Downtime
- Maintenance Alerts

Maintenance Queue

Work Order	Asset	Product	Type	Priority	Due	Status
MNT-1048	BIKE-017	Mountain Bike	Damage	High	Today	In Progress
MNT-1049	KAY-022	Kayak	Preventive	Medium	10 Sep	Scheduled
MNT-1050	BIKE-031	Mountain Bike	Corrective	Critical	Today	Awaiting Parts

Filters

- Venue
- Location
- Product
- Asset
- Maintenance Type
- Priority
- Technician
- Status
- Due Date

AI Alert

6 Mountain Bikes are forecast to require scheduled maintenance during next weekend's peak demand period.

Screen 2 — Maintenance Rule & Service Plan Configuration

Purpose

Define when different rental products/assets require preventive maintenance.

This is important because maintenance should not depend only on staff manually noticing a problem.

Maintenance Trigger Types

Calendar Based

Example:

Every 30 Days

Rental Count Based

Example:

Every 50 Rentals

Usage Based

Example:

Every 100 Rental Hours

Condition Based

Example:

Trigger inspection when condition becomes **Fair**.

Incident Based

Example:

Automatically create inspection after a safety-related incident.

Example — Mountain Bike

Preventive Service:

Every 50 rentals OR 100 rental hours OR 30 days
Whichever occurs first.

Service Plan

Configure:

- Product / Category
- Maintenance Type
- Trigger
- Service Checklist
- Estimated Duration
- Required Technician Skill
- Required Approval
- Post-Maintenance Inspection

Screen 3 — Maintenance Calendar & Scheduling

Purpose

Plan preventive and corrective maintenance while understanding its effect on rental capacity.

The original requirement specifically calls for scheduled maintenance periods.

Views

Day | Week | Month | Resource

Example

12 September

09:00–11:00

BIKE-017 — Brake Repair

10:00–12:00

BIKE-031 — Preventive Service

13:00–16:00

KAY-022 — Hull Inspection

Schedule Maintenance

Configure:

- Asset
- Maintenance Type
- Start Date/Time
- Expected Completion
- Technician
- Workshop/Location
- Priority

Critical Integration

Once scheduled:

Board 9 → Board 3

The affected equipment must automatically be removed from sellable availability for the maintenance period.

If future reservations are affected:

⚠ RESERVATION CONFLICT

TICVAI should alert the operator before confirming the maintenance schedule.

Screen 4 — Maintenance Work Order

Purpose

Create and manage the actual repair/service job.

Work Order

MNT-20260908-0045

Asset:

BIKE-017

Product:

Mountain Bike – Adult

Source

Return Damage Case

DMG-20260908-0021

Maintenance Type

Damage Repair

Priority

HIGH

Issue

Front brake lever bent during rental.

Work Order Information

- Created Date
- Created By
- Asset
- Location
- Maintenance Type
- Priority
- Assigned Technician
- Planned Start
- Expected Completion
- Workshop
- Estimated Cost

Status

Open → Assigned → In Progress → Awaiting Parts → Repair Complete → Inspection → Closed

Attachments

- Return photos
- Damage photos
- Incident records
- Technician photos
- Documents

Screen 5 — Technician Repair Workspace

Purpose

Provide the technician with a focused operational screen for executing maintenance.

Asset

BIKE-017

Work Order:

MNT-0045

Issue

Front brake lever damaged.

Maintenance Checklist

- ☒ Inspect brake lever
- ☒ Inspect brake cable
- ☐ Replace brake lever
- ☐ Test braking system

- ☐ Road/safety test
- ☐ Final inspection

Technician Actions

Start Work

Pause

Awaiting Parts

Complete Repair

Work Log

Time	Technician	Activity
09:05	Ahmed	Inspection started
09:18	Ahmed	Brake lever confirmed damaged
09:22	Ahmed	Replacement required

Notes

Brake cable remains in acceptable condition. Lever replacement required.

Evidence

Add Photo | Add Document

All actions must be timestamped.

Screen 6 — Parts, Cost & Maintenance Expense Tracking

Purpose

Capture the operational cost associated with maintaining rental equipment.

This should integrate with the broader Inventory/Procurement and Finance modules rather than recreate them.

Work Order

MNT-0045

Parts Used

Part	Qty	Unit Cost	Total
Brake Lever	1	AED 55	AED 55
Grip	1	AED 15	AED 15

Labor

Technician:

Ahmed

Time:

45 minutes

Internal Labor Cost:

AED 60

Total Maintenance Cost

Parts:

AED 70

Labor:

AED 60

External Service:

AED 0

TOTAL

AED 130

Source of Parts

Parts should be requested/consumed through TICVAI's Inventory & Procurement module where implemented.

Financial Treatment

Maintenance cost should be passed to Finance for accounting/reporting.

Screen 7 — Asset Maintenance History & Lifecycle

Purpose

Provide the complete maintenance history of an individual serialized rental asset.

The original requirement specifically requires maintenance history tracking.

Asset

BIKE-017

Purchased:

12 Jan 2026

Age:

8 Months

Total Rentals:

184

Total Rental Hours:

276

Total Revenue:

AED 11,840

Maintenance Summary

Preventive Services:

4

Corrective Repairs:

2

Damage Repairs:

3

Total Maintenance Cost:

AED 1,480

Total Downtime:

36 Hours

Timeline

12 Jan — Activated

↓

15 Mar — Preventive Maintenance

↓

04 May — Tire Repair

↓

18 Jul — Customer Damage

↓

08 Sep — Brake Repair

Asset Economics

I recommend showing:

Lifetime Revenue vs Maintenance Cost

This helps management decide when an asset is no longer economically sensible to maintain.

Screen 8 — Return-to-Service Inspection & Approval

Purpose

Ensure repaired equipment does not automatically become rentable when a technician clicks “Repair Complete.”

This is one of the most important controls in Board 9.

Repair

COMPLETE

does **not** mean:

AVAILABLE

Instead:

Repair Complete

↓

RETURN-TO-SERVICE INSPECTION

Inspection Checklist

For BIKE-017:

- ✓ Brake lever replaced
- ✓ Brake cable checked
- ✓ Brake function tested
- ✓ Tires checked
- ✓ Frame inspected
- ✓ Safety test completed

Inspection Result

- Pass
- Rework Required
- Unsafe

- Supervisor Review

Approved By

Supervisor:

Ahmed Khan

Date/Time:

12 Sep — 11:18

Result

✓ **SAFE FOR RENTAL**

Only then:

Maintenance → Turnaround → Available

Board 3 must then recalculate availability.

Screen 9 — Asset Retirement, Write-Off & Replacement Recommendation

Purpose

Manage assets that should no longer remain in the active rental fleet.

Retirement Reasons

- End of Useful Life
- Excessive Maintenance Cost
- Safety Concern
- Repeated Failure
- Total Loss
- Obsolete
- Lost / Not Recovered
- Management Decision

Asset

BIKE-031

Purchase Cost:

AED 2,500

Lifetime Revenue:

AED 8,920

Maintenance Cost:

AED 2,180

Repairs:

12

Downtime:

18 Days

Condition:

Poor

AI Recommendation

BIKE-031's maintenance cost has reached 87% of acquisition cost and its failure frequency is 3.2× the fleet average.

Recommendation

Consider Retirement / Replacement

Workflow

Recommendation → Review → Approval → Retire

Once retired:

- Asset removed permanently from availability.
- Existing future assignments must be identified.
- Asset retained in historical reporting.
- Financial/write-off process passed to Finance where applicable.

The physical asset record must never simply be deleted.

Screen 10 — Maintenance Intelligence & Predictive AI

Purpose

Use AI to move TICVAI from **reactive maintenance** toward **predictive maintenance**.

This is where I recommend making the module significantly stronger than the original matrix.

Maintenance Health Dashboard

Display:

- Assets at Risk
- Predicted Failures
- Upcoming Preventive Maintenance
- Repeat Failures
- High Maintenance Cost Assets
- Downtime by Product
- Maintenance Impact on Availability
- Mean Time Between Failures
- Mean Time to Repair

AI Predictive Maintenance

Example:

BIKE-018

94 rental hours since last service.

Recent inspection scores are declining.

Similar assets typically require brake servicing between 95–110 hours.

Recommendation: Schedule preventive maintenance within the next 10 rental hours.

AI Capacity-Aware Scheduling

This is particularly important.

Instead of simply saying:

Maintain 10 bikes Saturday.

AI should understand Board 3 demand.

Example:

Saturday utilization forecast: **94%**

Monday utilization forecast: **38%**

Recommended:

Schedule 8 preventive-maintenance jobs on Monday between 09:00–13:00 rather than Saturday.

AI Failure Pattern

7 of 12 recent repairs for Mountain Bike Model X involved the front braking system.

Recommendation:

Review maintenance frequency and supplier/component quality.

AI Replace vs Repair

Estimated next repair: AED 420

Current asset value: AED 650

Failure probability next 90 days: 71%

Recommendation: Replace rather than repair.

All recommendations remain advisory and subject to authorized approval.

Important Maintenance Trigger Architecture

I recommend Softlabs support multiple triggers simultaneously.

For example:

Mountain Bike Preventive Maintenance

Every 30 Days

OR

Every 50 Rentals

OR

Every 100 Rental Hours

OR

Condition = Fair

OR

Safety Incident

Whichever Happens First

This is much stronger than only scheduling maintenance by calendar date.

Important Maintenance → Availability Logic

The system must tightly integrate Board 9 with Board 3.

Example:

BIKE-017 scheduled maintenance:

12 Sep 10:00–14:00

Immediately:

Board 3

BIKE-017 becomes:

Unavailable 10:00–14:00 + required turnaround

If the maintenance finishes early:

Repair Complete at 12:30

↓

Return-to-Service Inspection at 12:45

↓

Approved at 12:50

Board 3 can recalculate availability from the approved return-to-service time.

If maintenance takes longer:

Expected completion changes:

14:00 → 17:00

Board 3 must immediately identify affected future reservations.

Important Maintenance Source Tracking

Every maintenance case should identify **why it exists**.

Possible source:

Preventive Schedule

Return Inspection

Damage Case

Active Rental Incident

Staff Inspection

Customer Complaint

AI Recommendation

Manufacturer Recall

This allows management to understand whether maintenance is primarily caused by:

- normal usage,
 - customer damage,
 - equipment quality,
 - operational handling,
 - or aging assets.
-

Important Asset Lifecycle

The full asset lifecycle should therefore become:

PROCURED

↓

RECEIVED

↓

ACTIVATED

↓

AVAILABLE

↓

Reserved

↓

Rented

↓

Returned

↓

Inspection

↓

Possible:

Available

or

MAINTENANCE

↓

Repair

↓

Return-to-Service Inspection

↓

AVAILABLE

Eventually:

RETIRED / WRITTEN OFF

The complete history must remain permanently traceable according to configured retention policies.

Board 9 — Core Maintenance Flow

Maintenance Trigger

↓

Create Work Order

↓

Remove From Availability



Schedule



Assign Technician



Inspection / Diagnosis



Repair / Service



Parts & Cost



Repair Complete



RETURN-TO-SERVICE INSPECTION



If Pass:

AVAILABLE

If Fail:

REWORK / OUT OF SERVICE



Update Asset History



AI Lifecycle Analysis

Integration Boundaries

Board 9 owns

- Rental equipment maintenance policies
- Preventive maintenance triggers
- Maintenance scheduling
- Rental asset work orders
- Technician workflow
- Repair/service history
- Return-to-service inspection
- Maintenance costs at asset/work-order level
- Asset retirement recommendation
- Predictive maintenance intelligence

Board 2 owns

Asset master, serial number, physical location and general asset status.

Board 3 owns

Rental availability and capacity impact.

Board 7 owns

Incidents occurring during an active rental.

Board 8 owns

Return inspection, customer damage assessment and settlement.

Inventory & Procurement owns

Parts inventory, purchase orders and supplier procurement.

Finance owns

Financial posting, write-offs and accounting treatment.

Board 10 will own

Consolidated rental management reporting, analytics and executive AI intelligence.

Board 9 — Acceptance Criteria

Board 9 is complete when TICVAI can:

1. Display a centralized maintenance command center for the rental fleet.
2. Configure preventive maintenance by **calendar, rental count, usage hours, condition and incidents**.
3. Automatically create maintenance requirements from qualifying triggers.
4. Schedule preventive, corrective and damage-repair maintenance as required by the original scope.
5. Automatically remove equipment from rental availability during maintenance.
6. Detect future reservation conflicts created by maintenance.
7. Create and manage structured maintenance work orders.
8. Assign technicians, priorities, workshops and expected completion dates.
9. Provide technicians with structured service/repair checklists.
10. Capture technician notes, timestamps, photos and documents.
11. Support **Open, Assigned, In Progress, Awaiting Parts, Repair Complete, Inspection and Closed** statuses.
12. Track parts, labor, external service and total maintenance cost.
13. Integrate parts consumption with Inventory & Procurement rather than duplicating stock management.
14. Maintain complete maintenance history by serialized asset as required by the source.
15. Track rental count, usage hours, downtime and lifetime maintenance cost.
16. Require a controlled **return-to-service inspection** before repaired equipment becomes rentable again.
17. Immediately recalculate availability after approved return to service.
18. Support rework and unsafe/out-of-service outcomes when inspection fails.
19. Support controlled asset retirement/write-off workflows without deleting historical records.
20. Compare lifetime revenue, maintenance cost, downtime and failure frequency at asset level.
21. Use AI to predict maintenance requirements and recurring equipment failures.
22. Use AI and Board 3 demand forecasts to recommend the **best maintenance window** with the least revenue/capacity impact.
23. Provide AI-assisted **repair-versus-replace recommendations**.
24. Maintain a complete audit trail from **maintenance trigger → work order → repair → inspection → return to service/retirement**.

This makes Board 9 the **health-management system for the rental fleet**: TICVAI doesn't only know whether an asset is “available” or “under maintenance”; it understands **why it failed, what was done to it, what it cost, when it is safe to return, when it is likely to fail again, and when replacing it becomes smarter than repairing it**.

Board 10 — Rental Command Center, Reporting, Analytics & AI Intelligence



Module: 7 — Rental Management

Board Objective: Provide management, operations, finance, maintenance, and venue teams with a **single management and intelligence layer for the entire Rental Management module**, consolidating operational KPIs, utilization, revenue, inventory performance, customer behavior, maintenance impact, exceptions, audit history, forecasting, and AI recommendations.

The original requirements explicitly call for rental reporting covering **rental revenue, utilization rate, rental duration, late returns, damaged items, deposit collection/refund, inventory availability and equipment performance**, with export to **Excel, PDF and CSV**. They also require role-based access and full audit logging.

The source additionally recommends **AI-based utilization and demand forecasting**.

However, for TICVAI I recommend making Board 10 much stronger than a traditional reporting dashboard.

Board 10 should answer:

“How is the rental business performing, what is happening operationally, what is likely to happen next, and what should management do about it?”

I recommend exactly **10 screens**.

Screen 1 — Rental Executive Command Center

Purpose

Provide senior management with an immediate view of the overall rental business.

Global Filters

- Tenant
- Venue
- Rental Location
- Product

- Category
- Date Range
- Channel
- Customer Segment

Executive KPI Cards

Rental Revenue

AED 1.84M

▲ 12.4%

Total Rentals

24,860

▲ 8.2%

Average Utilization

74%

▲ 5.1%

Average Rental Value

AED 74

On-Time Return Rate

91%

Equipment Availability

87%

Damage Rate

2.8%

Maintenance Downtime

4.3%

Business Performance

Charts:

Revenue Trend

Rentals Trend

Utilization Trend

Today

- Active Rentals
- Upcoming Reservations
- Due Soon
- Overdue
- Maintenance Blocked
- Equipment Available

AI Executive Summary

Rental revenue is 12.4% above the previous comparable period, primarily driven by higher weekend kayak utilization and increased average bicycle rental duration.

Mountain Bike availability at North Station is expected to become constrained this weekend.

Screen 2 — Rental Revenue & Commercial Analytics

Purpose

Analyze rental commercial performance.

The original scope specifically requires reporting on rental revenue.

KPIs

- Gross Rental Revenue
- Net Rental Revenue
- Average Transaction Value
- Average Revenue per Rental Hour
- Extension Revenue
- Late-Fee Revenue
- Damage Charges
- Deposit Captures
- Discount Value
- Refund Value

Revenue Breakdown

Analyze by:

- Product
- Category
- Venue
- Location
- Channel
- Day
- Time
- Customer segment

Example

Product	Rentals	Revenue	Avg. Value	Rev./Hour
Mountain Bike	8,420	AED 589K	AED 70	AED 48
Double Kayak	4,210	AED 631K	AED 150	AED 92
Stroller	7,800	AED 312K	AED 40	AED 35
Cabana	1,220	AED 427K	AED 350	AED 67

AI Commercial Insight

120-minute bicycle rentals consume twice the capacity of 60-minute rentals but generate only 58% additional revenue.

Recommendation:

Review the 120-minute pricing tier.

This should connect management back to **Board 4 Pricing**.

Screen 3 — Utilization & Capacity Analytics

Purpose

Understand how efficiently rental inventory is being used.

The original requirements specifically call for **utilization-rate reporting**.

KPIs

- Overall Utilization
- Peak Utilization
- Off-Peak Utilization
- Idle Inventory
- Sold-Out Hours
- Lost Sales Due to Capacity
- Protected Inventory Utilization

Utilization Heatmap

Example:

Time	Mon	Tue	Wed	Thu	Fri	Sat	Sun
09–12	42%	39%	44%	48%	61%	78%	74%
12–15	55%	51%	58%	63%	76%	92%	89%
15–18	62%	60%	65%	72%	88%	97%	94%

Analyze By

Product | Location | Date | Time | Duration

AI Insight

Saturday bicycle utilization reaches 97% between 15:00–18:00 while Monday morning utilization averages only 31%.

Recommendation:

Shift maintenance and non-essential inventory activity to Monday mornings.

Screen 4 — Inventory & Equipment Performance Analytics

Purpose

Measure the operational performance of the physical rental fleet.

The original requirements call for reporting on **inventory availability and equipment performance**.

Fleet KPIs

- Total Assets
- Available
- Rented
- Maintenance
- Out of Service
- Lost
- Average Rentals per Asset
- Average Revenue per Asset

Equipment Performance

Asset	Rentals	Hours	Revenue	Damage	Downtime	Health
BIKE-017	184	276	AED 11,840	3	36h	Good
BIKE-018	201	302	AED 13,220	1	14h	Excellent
BIKE-031	96	140	AED 8,920	7	18d	Poor

Identify

- Most utilized assets
- Least utilized assets
- Highest revenue assets
- Highest maintenance assets
- Repeatedly damaged assets
- Excessively idle inventory

AI Recommendation

BIKE-031 has a high downtime and repair frequency relative to fleet peers.

Recommendation: Review for replacement.

Screen 5 — Rental Duration, Extension & Return Analytics

Purpose

Understand customer usage behavior after checkout.

The source explicitly requires reporting on rental duration and late returns.

KPIs

- Average Planned Duration
- Average Actual Duration
- Extension Rate
- Average Extension
- On-Time Return %
- Late Return %
- Average Late Duration
- Critical Overdue Count

Duration Distribution

30 Minutes — 12%

60 Minutes — 38%

90 Minutes — 24%

120 Minutes — 18%

2+ Hours — 8%

Extension Analytics

Extension Requests:

1,420

Approved:

1,086

Declined Due to Capacity:

334

Extension Revenue:

AED 91K

AI Insight

31% of Saturday kayak customers request an extension.

Recommendation:

Consider introducing a 150-minute rental option or revising standard duration packages.

Screen 6 — Damage, Loss, Deposit & Exception Analytics

Purpose

Give management visibility into operational and financial risk.

The original scope explicitly requires reporting on **damaged items and deposit collection/refund**.

KPIs

- Damage Cases
- Damage Rate
- Lost Equipment
- Total Damage Charges
- Deposit Collected/Authorized
- Deposit Released
- Deposit Captured
- Deposit Pending
- Waived Fees
- Manual Overrides

Damage Breakdown

By:

- Product
- Asset
- Location
- Damage category
- Customer responsibility
- Time period

Example

Damage Cause

Customer Damage — 42%

Normal Wear — 31%

Equipment Failure — 18%

Undetermined — 9%

Deposit Analytics

Authorized:

AED 4.2M

Released:
AED 3.96M
Partially Captured:
AED 182K
Fully Captured:
AED 58K

AI Risk Insight

Damage frequency for Mountain Bike Model X is 2.3× the bicycle fleet average.
Recommendation:
Review equipment model quality, maintenance frequency and customer safety briefing.

Screen 7 — Location & Channel Performance

Purpose

Compare rental performance across physical locations and sales channels.

Location Comparison

Location	Rentals	Revenue	Utilization	Availability	Damage
North Station	9,420	AED 710K	88%	79%	3.2%
Marina A	7,810	AED 840K	82%	84%	2.1%
Main Gate	5,100	AED 310K	61%	93%	1.4%

Channel Comparison

- B2C
- B2B
- POS
- Walk-In
- Mobile App
- API / Partner

Analyze:

Bookings

Revenue

Average Value

Cancellation

No-Show

Utilization Contribution

AI Rebalancing Insight

Marina A has 17% excess weekday bicycle inventory while North Station experiences repeated sell-outs.
Recommendation:
Reallocate 8 bicycles from Marina A to North Station Monday–Friday.
This connects directly with Board 2 inventory transfers.

Screen 8 — Rental Forecasting & Demand Intelligence

Purpose

Forecast future rental demand and identify capacity problems before they happen.

This formally incorporates the AI utilization/demand forecasting recommendation from the source.

Forecast Horizon

Today | 7 Days | 30 Days | 90 Days

Forecast KPIs

- Expected Rentals
- Expected Revenue
- Forecast Utilization
- Sell-Out Probability
- Inventory Shortage Risk
- Maintenance Capacity Impact

Example

Saturday — North Station

Mountain Bikes

Physical Fleet:

48

Expected Rentable:

41

Forecast Peak Demand:

52

SHORTAGE: 11

Sell-Out Probability:

94%

AI Recommended Actions

1. Transfer 8 bikes from Marina.
2. Delay 4 non-critical maintenance jobs until Monday.
3. Release unused B2B allocation 90 minutes before slot.
4. Review peak pricing for 15:00–18:00.

This is where TICVAI's AI layer becomes particularly valuable because it can combine information from **Inventory + Availability + Pricing + Reservations + Maintenance**.

Screen 9 — Audit, Governance & Operational Control

Purpose

Provide management and administrators with traceability across Rental Management.

The source requires role-based permissions and audit logging for rental operations.

Audit Events

Track:

- Product configuration change
- Inventory adjustment
- Asset status override
- Price change
- Deposit override
- Booking modification
- Cancellation
- Equipment assignment
- Rental extension
- Equipment swap
- Late-fee waiver
- Damage assessment
- Damage-fee override
- Deposit capture
- Maintenance status change
- Asset retirement

Audit Table

Date/Time	User	Action	Entity	Before	After	Approval
08 Sep 14:22	Ahmed	Late Fee Waiver	RNT-10482	AED 80	AED 0	Manager
08 Sep 15:10	Omar	Asset Status	BIKE-017	Rented	Inspection	Auto
08 Sep 16:42	Admin	Pricing	Kayak	AED 120	AED 135	Approved

Search

Search by:

Rental ID | Customer | Asset | User | Action | Date

Export

Excel | CSV | PDF

as required by the source.

Role-Based Visibility

For example:

Rental Operator

Operational rental data only.

Maintenance

Asset and maintenance information.

Finance

Commercial/deposit data.

Manager

Location-level analytics.

Administrator

Full configuration and audit access.

Screen 10 — AI Rental Management Copilot & Action Center

Purpose

Provide a natural-language AI interface across the complete Rental Management module.

I strongly recommend this as the final screen because it aligns the rental module with TICVAI's overall **AI-first architecture**.

Instead of management having to manually navigate every report, they should be able to ask TICVAI questions.

AI Query

“Why did rental revenue drop yesterday?”

TICVAI AI

Rental revenue decreased 8.4% compared with the previous Tuesday.

Primary contributors:

- 1. North Station bicycle availability declined 14% because 7 assets were under maintenance.**
- 2. Kayak bookings were 11% lower between 14:00–17:00.**
- 3. 6 cabana reservations were cancelled because of the weather closure.**

Estimated revenue impact:

AED 18,420

Another Query

“Which equipment should I replace?”

AI response:

Highest replacement priority

1. BIKE-031
2. BIKE-044
3. KAY-012

Reason:

- High maintenance cost
 - Repeated failures
 - Excessive downtime
 - Lower utilization
 - Poor asset health
-

Another Query

“What should I do for this weekend?”

AI response:

Forecast demand indicates a high risk of bicycle and kayak shortages Saturday between 15:00–19:00.

Recommended actions:

Transfer 8 bicycles → North Station

Move 4 preventive maintenance jobs → Monday

Release unused B2B inventory 90 minutes before start

Consider +10% kayak peak pricing within configured guardrails

Actionable AI

Where the user has permission, each recommendation can provide:

Review Transfer

Open Pricing

Reschedule Maintenance

Review Availability

Create Report

AI should **not directly execute sensitive commercial or inventory actions without the required user authorization and approval workflow.**

Important Recommendation — Reporting vs Power BI

Because TICVAI already has broader reporting/BI capabilities, I would **not build an independent BI platform inside Rental Management.**

Board 10 should provide:

Operational Rental Dashboards

for daily management.

and expose its governed data to:

TICVAI Reporting / Power BI / AI Analytics Layer

for deeper enterprise analytics.

This avoids duplicating the Reporting module while still giving Rental users the dashboards they need.

Important Recommendation — One Rental Data Model

Board 10 should not generate KPIs from separate copies of information.

It should consume the same underlying rental lifecycle:

Product

- **Inventory**
- **Availability**
- **Pricing**
- **Reservation**
- **Checkout**
- **Active Rental**
- **Return**
- **Settlement**
- **Maintenance**

That allows management to drill from:

Revenue = AED 1.84M

down to:

Venue

- **Location**
- **Product**
- **Rental**
- **Customer Transaction**
- **Physical Asset**

This drill-down capability will be extremely valuable.

Important Recommendation — AI Should Explain, Not Only Predict

For TICVAI, I recommend every major AI recommendation include:

Observation

What happened?

Cause

Why?

Prediction

What is likely to happen?

Recommendation

What should we do?

Expected Impact

What happens if we do it?

Example:

Observation: North Station bike availability is falling.

Cause: 7 assets are under maintenance and weekend demand is 18% above normal.

Prediction: 94% probability of sell-out Saturday 15:00–18:00.

Recommendation: Transfer 8 bikes from Marina A.

Expected Impact: Reduce projected lost rental opportunities from 37 to approximately 9.

That is much more valuable than simply displaying:

“High demand expected.”

Board 10 — Management Flow

Rental Operational Data

From Boards 1–9



CONSOLIDATED RENTAL DATA MODEL



KPIs & Dashboards



Operational Analytics



Commercial Analytics



Asset & Maintenance Analytics



Forecasting



TICVAI AI INTELLIGENCE



Observation



Explanation



Prediction



Recommendation



Authorized Action



MEASURE RESULT

This closes the AI feedback loop.

Integration Boundaries

Board 10 owns

- Rental-specific management dashboards
- Rental operational KPIs
- Rental utilization analytics
- Commercial performance analytics
- Asset performance analytics
- Return/damage/deposit analytics
- Rental demand forecasting
- Rental-specific AI recommendations
- Rental audit exploration
- Natural-language rental analytics

Enterprise Reporting / Power BI owns

- Cross-module BI
- Enterprise dashboards
- Advanced reporting models
- Corporate executive reporting

Finance owns

- Accounting
- Revenue recognition
- Financial statements
- Settlement/reconciliation

CRM owns

- Broader customer segmentation and marketing analytics

Dynamic Pricing owns

- Central pricing models and advanced pricing experimentation

AI Layer owns

- Shared LLM/model infrastructure
- AI governance
- Model usage
- Security
- Consumption monitoring

Board 10 consumes those shared services for **Rental Management intelligence**.

Board 10 — Acceptance Criteria

Board 10 is complete when TICVAI can:

1. Provide an executive Rental Management command center.
2. Report rental revenue and commercial performance.
3. Report utilization by product, asset, location, date and time.
4. Analyze planned versus actual rental duration.
5. Analyze extensions and extension revenue.
6. Report late returns and overdue behavior.
7. Analyze damage, loss and equipment incidents.
8. Report deposit authorization, release, partial capture and forfeiture.
9. Measure physical asset performance and revenue per asset.
10. Measure maintenance cost, downtime and fleet health.
11. Compare rental locations and sales channels.
12. Identify idle, underutilized and capacity-constrained inventory.
13. Forecast future rental demand and utilization.
14. Predict inventory shortages and sell-out risks.
15. Combine demand forecasts with maintenance schedules.
16. Recommend inventory transfers and rebalancing.
17. Recommend pricing review opportunities without duplicating the Dynamic Pricing engine.
18. Recommend asset maintenance and replacement opportunities.
19. Provide complete rental audit search and drill-down.
20. Enforce role-based access to dashboards and sensitive data.
21. Export applicable reports to **Excel, CSV and PDF** as required by the source.
22. Allow drill-down from executive KPI to transaction and serialized asset where permissions allow.
23. Provide natural-language AI querying of rental information.
24. Explain **what happened, why it happened, what is likely to happen, and what action is recommended**.
25. Estimate the expected operational or commercial impact of AI recommendations.
26. Route recommended actions back to the appropriate operational board rather than duplicating those functions.

27. Maintain auditability of AI recommendations, user decisions and resulting actions.

Final Rental Module Architecture

With Board 10, the complete module is now:

- Board 1 — Rental Product Configuration**
- Board 2 — Rental Inventory & Equipment**
- Board 3 — Availability & Capacity Engine**
- Board 4 — Pricing, Deposits & Commercial Rules**
- Board 5 — Booking & Reservation Management**
- Board 6 — Checkout & Equipment Assignment**
- Board 7 — Active Rental Operations**
- Board 8 — Return, Damage & Settlement**
- Board 9 — Maintenance & Asset Lifecycle**
- Board 10 — Command Center, Reporting & AI Intelligence**

This gives us a complete **end-to-end Rental Management architecture** rather than simply implementing the original requirements as isolated features. Board 10 closes the loop by turning everything generated by Boards 1–9 into **management intelligence, forecasts, AI recommendations, and measurable actions**.